

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 1                 |        |        |        |        |        |              |
| DLT rate                   | 0.30   | 0.35   | 0.40   | 0.45   | 0.50   |              |
| Efficacy rate              | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |              |
| Utility 2                  | -0.2   | -0.3   | -0.3   | -0.4   | -0.4   |              |
| Utility 3                  | 31     | 29.6   | 28.2   | 26.8   | 25.4   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 68     | 13.6   | 4.7    | 1.7    | 0      | 12           |
| U2 - dose-specific alpha   | 67.7   | 13.6   | 5      | 1.7    | 0      | 12           |
| U3 - shared alpha          | 68.3   | 13     | 4.9    | 1.7    | 0      | 12.1         |
| U3 - dose-specific alpha   | 68.1   | 13.4   | 4.8    | 1.6    | 0.1    | 12           |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 14.7   | 7.2    | 1      | 0.2    | 0      | 76.9         |
| U2 - dose-specific alpha   | 14.8   | 6.9    | 1.4    | 0.1    | 0      | 76.8         |
| U3 - shared alpha          | 15.8   | 6.2    | 1.1    | 0.1    | 0      | 76.8         |
| U3 - dose-specific alpha   | 16.1   | 5.6    | 1      | 0.2    | 0      | 77.1         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 11.3   | 3.3    | 1.7    | 0.6    | 0.2    | -            |
| U2 - dose-specific alpha   | 11.3   | 3.3    | 1.7    | 0.6    | 0.2    | -            |
| U3 - shared alpha          | 11.3   | 3.3    | 1.7    | 0.6    | 0.2    | -            |
| U3 - dose-specific alpha   | 11.3   | 3.4    | 1.7    | 0.6    | 0.2    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 3.3    | 1.7    | 0.6    | 0.3    | 0.1    | -            |
| U2 - dose-specific alpha   | 3.3    | 1.7    | 0.6    | 0.3    | 0.1    | -            |
| U3 - shared alpha          | 3.3    | 1.7    | 0.6    | 0.3    | 0.1    | -            |
| U3 - dose-specific alpha   | 3.3    | 1.7    | 0.6    | 0.3    | 0.1    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 607.9        |
| U2 - dose-specific alpha   |        |        |        |        |        | 607.7        |
| U3 - shared alpha          |        |        |        |        |        | 608.2        |
| U3 - dose-specific alpha   |        |        |        |        |        | 608.4        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 2                 |      |      |      |      |      |        |
| DLT rate                   | 0.15 | 0.20 | 0.30 | 0.35 | 0.45 |        |
| Efficacy rate              | 0.20 | 0.25 | 0.50 | 0.50 | 0.50 |        |
| Utility 2                  | 0.1  | 0.1  | 0.2  | 0.1  | 0.1  |        |
| Utility 3                  | 46   | 47   | 58   | 56   | 52   |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 24.7 | 20.9 | 30.2 | 17.1 | 6.6  | 0.5    |
| U2 - dose-specific alpha   | 24.6 | 21.2 | 30   | 17.1 | 6.6  | 0.5    |
| U3 - shared alpha          | 24.5 | 21.5 | 29.1 | 18.2 | 6.2  | 0.5    |
| U3 - dose-specific alpha   | 24.7 | 20.8 | 30.6 | 17   | 6.4  | 0.5    |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 34.3 | 17.3 | 31.6 | 9.9  | 2.1  | 4.8    |
| U2 - dose-specific alpha   | 33.9 | 17.9 | 31.5 | 9.4  | 2.3  | 5      |
| U3 - shared alpha          | 36.9 | 18.6 | 29.4 | 8.9  | 1.3  | 4.9    |
| U3 - dose-specific alpha   | 37.2 | 18.4 | 30.2 | 8    | 1.4  | 4.8    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 11.5 | 5.3  | 7.1  | 3.6  | 1.6  | -      |
| U2 - dose-specific alpha   | 11.5 | 5.4  | 7    | 3.6  | 1.6  | -      |
| U3 - shared alpha          | 11.6 | 5.3  | 7.1  | 3.7  | 1.6  | -      |
| U3 - dose-specific alpha   | 11.5 | 5.3  | 7.1  | 3.6  | 1.6  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 3.3  | 2.5  | 2    | 1    | 0.4  | -      |
| U2 - dose-specific alpha   | 3.3  | 2.5  | 1.9  | 1    | 0.4  | -      |
| U3 - shared alpha          | 3.4  | 2.5  | 1.9  | 1    | 0.4  | -      |
| U3 - dose-specific alpha   | 3.3  | 2.5  | 2    | 1    | 0.4  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1098.9 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1098.1 |
| U3 - shared alpha          |      |      |      |      |      | 1099.3 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1099   |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 3                 |        |        |        |        |        |              |
| DLT rate                   | 0.15   | 0.20   | 0.30   | 0.35   | 0.45   |              |
| Efficacy rate              | 0.10   | 0.40   | 0.40   | 0.40   | 0.40   |              |
| Utility 2                  | -0.1   | 0.2    | 0.1    | 0.1    | -0.1   |              |
| Utility 3                  | 40     | 56     | 52     | 50     | 46     |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 25.7   | 20.9   | 28.9   | 17.5   | 6.4    | 0.6          |
| U2 - dose-specific alpha   | 25.6   | 21     | 29.2   | 17.5   | 6.1    | 0.6          |
| U3 - shared alpha          | 26     | 20.4   | 29.4   | 17.4   | 6.2    | 0.6          |
| U3 - dose-specific alpha   | 25.5   | 21     | 30     | 16.3   | 6.6    | 0.6          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 12.7   | 41.7   | 18.3   | 6.9    | 1.9    | 18.5         |
| U2 - dose-specific alpha   | 12.6   | 41.1   | 18.6   | 7.8    | 1.5    | 18.4         |
| U3 - shared alpha          | 13.7   | 43.3   | 17     | 6.1    | 1.3    | 18.6         |
| U3 - dose-specific alpha   | 13     | 43.9   | 16.9   | 6.2    | 1.5    | 18.5         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 8.6    | 7.7    | 5.9    | 3.3    | 1.5    | -            |
| U2 - dose-specific alpha   | 8.6    | 7.7    | 5.9    | 3.4    | 1.5    | -            |
| U3 - shared alpha          | 8.6    | 7.7    | 5.8    | 3.4    | 1.5    | -            |
| U3 - dose-specific alpha   | 8.6    | 7.7    | 6      | 3.3    | 1.5    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 2.3    | 3.3    | 1.7    | 1.1    | 0.4    | -            |
| U2 - dose-specific alpha   | 2.3    | 3.3    | 1.6    | 1.1    | 0.4    | -            |
| U3 - shared alpha          | 2.3    | 3.3    | 1.6    | 1.1    | 0.4    | -            |
| U3 - dose-specific alpha   | 2.3    | 3.3    | 1.7    | 1      | 0.5    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1004.8       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1005.3       |
| U3 - shared alpha          |        |        |        |        |        | 1005         |
| U3 - dose-specific alpha   |        |        |        |        |        | 1005.1       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 4                 |      |      |      |      |      |        |
| DLT rate                   | 0.05 | 0.10 | 0.18 | 0.30 | 0.40 |        |
| Efficacy rate              | 0.05 | 0.10 | 0.25 | 0.50 | 0.55 |        |
| Utility 2                  | 0    | 0    | 0.1  | 0.2  | 0.1  |        |
| Utility 3                  | 41   | 42   | 47.8 | 58   | 57   |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 2.7  | 3.2  | 24   | 42.7 | 27.4 | 0      |
| U2 - dose-specific alpha   | 2.7  | 3.5  | 23.5 | 43   | 27.3 | 0      |
| U3 - shared alpha          | 2.7  | 3.4  | 24.6 | 41.8 | 27.5 | 0      |
| U3 - dose-specific alpha   | 2.7  | 3.4  | 24   | 42.6 | 27.3 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 7.2  | 7.6  | 23.4 | 42.8 | 15.4 | 3.6    |
| U2 - dose-specific alpha   | 7.2  | 8.5  | 22.8 | 43.6 | 14.3 | 3.6    |
| U3 - shared alpha          | 11.1 | 9.2  | 21.4 | 41.2 | 13.5 | 3.6    |
| U3 - dose-specific alpha   | 11.3 | 9.1  | 21.8 | 42   | 12.3 | 3.5    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 4.4  | 4.3  | 6.8  | 8.1  | 5.8  | -      |
| U2 - dose-specific alpha   | 4.4  | 4.3  | 6.7  | 8.2  | 5.8  | -      |
| U3 - shared alpha          | 4.4  | 4.3  | 6.7  | 8.1  | 5.8  | -      |
| U3 - dose-specific alpha   | 4.4  | 4.4  | 6.7  | 8.1  | 5.8  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.6  | 2.8  | 2.1  | 2.7  | 1.4  | -      |
| U2 - dose-specific alpha   | 0.6  | 2.8  | 2.1  | 2.7  | 1.4  | -      |
| U3 - shared alpha          | 0.7  | 2.8  | 2.1  | 2.7  | 1.4  | -      |
| U3 - dose-specific alpha   | 0.6  | 2.8  | 2.1  | 2.6  | 1.4  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1089.2 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1089.3 |
| U3 - shared alpha          |      |      |      |      |      | 1088.5 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1090.2 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 5                 |        |        |        |        |        |              |
| DLT rate                   | 0.07   | 0.12   | 0.17   | 0.22   | 0.30   |              |
| Efficacy rate              | 0.05   | 0.30   | 0.30   | 0.30   | 0.30   |              |
| Utility 2                  | -0     | 0.2    | 0.1    | 0.1    | 0      |              |
| Utility 3                  | 40.2   | 53.2   | 51.2   | 49.2   | 46     |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 6.2    | 3.9    | 14.4   | 26.7   | 48.7   | 0.1          |
| U2 - dose-specific alpha   | 6.2    | 3.9    | 14.3   | 27.1   | 48.4   | 0.1          |
| U3 - shared alpha          | 6.2    | 4.1    | 14     | 26.9   | 48.7   | 0.1          |
| U3 - dose-specific alpha   | 6.2    | 3.9    | 14.8   | 25.6   | 49.4   | 0.1          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 3.9    | 36.4   | 25.8   | 18.4   | 9.4    | 6.1          |
| U2 - dose-specific alpha   | 3.2    | 35.4   | 25.4   | 20     | 9.9    | 6.1          |
| U3 - shared alpha          | 6.8    | 41.1   | 24.5   | 16.3   | 5.2    | 6.1          |
| U3 - dose-specific alpha   | 6.2    | 41.2   | 25     | 15.9   | 5.6    | 6.1          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 4.6    | 6.6    | 6.7    | 5.8    | 5.2    | -            |
| U2 - dose-specific alpha   | 4.6    | 6.6    | 6.7    | 5.8    | 5.2    | -            |
| U3 - shared alpha          | 4.5    | 6.6    | 6.7    | 5.8    | 5.1    | -            |
| U3 - dose-specific alpha   | 4.5    | 6.5    | 6.8    | 5.8    | 5.2    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 0.7    | 3.5    | 2.1    | 2.2    | 1.8    | -            |
| U2 - dose-specific alpha   | 0.7    | 3.5    | 2.1    | 2.2    | 1.8    | -            |
| U3 - shared alpha          | 0.7    | 3.5    | 2.2    | 2.2    | 1.8    | -            |
| U3 - dose-specific alpha   | 0.7    | 3.5    | 2.2    | 2.2    | 1.8    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1014.1       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1014.5       |
| U3 - shared alpha          |        |        |        |        |        | 1014.8       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1015.3       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 6                 |      |      |      |      |      |        |
| DLT rate                   | 0.29 | 0.34 | 0.38 | 0.43 | 0.52 |        |
| Efficacy rate              | 0.30 | 0.32 | 0.35 | 0.38 | 0.40 |        |
| Utility 2                  | 0    | -0   | -0   | -0.1 | -0.1 |        |
| Utility 3                  | 46.4 | 45.6 | 45.8 | 45.6 | 43.2 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 59.5 | 16   | 7.2  | 2.4  | 0.3  | 14.6   |
| U2 - dose-specific alpha   | 59.1 | 16.7 | 6.3  | 2.7  | 0.4  | 14.8   |
| U3 - shared alpha          | 59.5 | 16.3 | 6.7  | 2.4  | 0.4  | 14.7   |
| U3 - dose-specific alpha   | 59.6 | 16.3 | 6.4  | 2.7  | 0.3  | 14.7   |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 66.6 | 11.3 | 3    | 0.6  | 0.2  | 18.3   |
| U2 - dose-specific alpha   | 66.7 | 11.5 | 2.8  | 0.8  | 0.3  | 17.9   |
| U3 - shared alpha          | 67.9 | 11.2 | 2.5  | 0.2  | 0.2  | 18     |
| U3 - dose-specific alpha   | 67.2 | 11.7 | 2.6  | 0.5  | 0.1  | 17.9   |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 19.6 | 3.8  | 2.2  | 0.8  | 0.2  | -      |
| U2 - dose-specific alpha   | 19.7 | 3.8  | 2.2  | 0.8  | 0.3  | -      |
| U3 - shared alpha          | 19.7 | 3.8  | 2.1  | 0.9  | 0.3  | -      |
| U3 - dose-specific alpha   | 19.7 | 3.9  | 2.1  | 0.9  | 0.3  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 5.4  | 1.5  | 0.6  | 0.3  | 0.1  | -      |
| U2 - dose-specific alpha   | 5.4  | 1.5  | 0.6  | 0.3  | 0.1  | -      |
| U3 - shared alpha          | 5.4  | 1.5  | 0.6  | 0.3  | 0.1  | -      |
| U3 - dose-specific alpha   | 5.4  | 1.5  | 0.6  | 0.3  | 0.1  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1034.5 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1037.8 |
| U3 - shared alpha          |      |      |      |      |      | 1037   |
| U3 - dose-specific alpha   |      |      |      |      |      | 1038.3 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 7                 |        |        |        |        |        |              |
| DLT rate                   | 0.16   | 0.20   | 0.26   | 0.31   | 0.36   |              |
| Efficacy rate              | 0.20   | 0.50   | 0.50   | 0.50   | 0.50   |              |
| Utility 2                  | 0      | 0.3    | 0.2    | 0.2    | 0.1    |              |
| Utility 3                  | 45.6   | 62     | 59.6   | 57.6   | 55.6   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 26     | 16.8   | 22.5   | 22.4   | 11.6   | 0.7          |
| U2 - dose-specific alpha   | 26     | 16.6   | 22.9   | 22     | 11.8   | 0.7          |
| U3 - shared alpha          | 26     | 17.1   | 23     | 21.1   | 12.1   | 0.7          |
| U3 - dose-specific alpha   | 26     | 16.6   | 23.3   | 21.6   | 11.8   | 0.7          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 25.2   | 36.8   | 19.6   | 10.6   | 2.2    | 5.6          |
| U2 - dose-specific alpha   | 25.1   | 38     | 19.6   | 9.6    | 2.1    | 5.6          |
| U3 - shared alpha          | 27.4   | 39.8   | 17     | 8.8    | 1.4    | 5.6          |
| U3 - dose-specific alpha   | 26.8   | 41.1   | 16.6   | 8.4    | 1.6    | 5.5          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 10.9   | 7      | 5.5    | 3.7    | 1.9    | -            |
| U2 - dose-specific alpha   | 10.9   | 7      | 5.5    | 3.6    | 2      | -            |
| U3 - shared alpha          | 10.8   | 7.1    | 5.5    | 3.6    | 2      | -            |
| U3 - dose-specific alpha   | 10.9   | 7.1    | 5.5    | 3.6    | 1.9    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 3      | 2.8    | 1.3    | 1      | 0.5    | -            |
| U2 - dose-specific alpha   | 3.1    | 2.8    | 1.3    | 1      | 0.5    | -            |
| U3 - shared alpha          | 3      | 2.8    | 1.3    | 1      | 0.5    | -            |
| U3 - dose-specific alpha   | 3      | 2.9    | 1.4    | 1      | 0.5    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1114.9       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1113.9       |
| U3 - shared alpha          |        |        |        |        |        | 1114.4       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1114.3       |

|                            |      |      |      |      |      |       |
|----------------------------|------|------|------|------|------|-------|
| Scenario 8                 |      |      |      |      |      |       |
| DLT rate                   | 0.32 | 0.37 | 0.42 | 0.48 | 0.52 |       |
| Efficacy rate              | 0.10 | 0.15 | 0.20 | 0.30 | 0.40 |       |
| Utility 2                  | -0.2 | -0.2 | -0.2 | -0.2 | -0.1 |       |
| Utility 3                  | 33.2 | 34.2 | 35.2 | 38.8 | 43.2 |       |
| MTD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 67.6 | 9.2  | 4    | 1    | 0.2  | 18    |
| U2 - dose-specific alpha   | 66.5 | 9.5  | 4.1  | 1.2  | 0.1  | 18.6  |
| U3 - shared alpha          | 67.6 | 9.3  | 4    | 0.9  | 0.2  | 18    |
| U3 - dose-specific alpha   | 67.4 | 9.1  | 4.1  | 1.1  | 0.2  | 18.1  |
| OBD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 28.7 | 5.7  | 1.9  | 0.7  | 0.2  | 62.8  |
| U2 - dose-specific alpha   | 27.8 | 6.5  | 1.8  | 0.9  | 0.1  | 62.9  |
| U3 - shared alpha          | 29.8 | 5.4  | 1.2  | 0.5  | 0.2  | 62.9  |
| U3 - dose-specific alpha   | 29.4 | 5.1  | 1.5  | 0.7  | 0.2  | 63.1  |
| Mean administrations       |      |      |      |      |      |       |
| U2 - shared alpha          | 14   | 3.3  | 1.5  | 0.4  | 0.1  | -     |
| U2 - dose-specific alpha   | 14   | 3.3  | 1.5  | 0.5  | 0.1  | -     |
| U3 - shared alpha          | 14   | 3.3  | 1.5  | 0.4  | 0.1  | -     |
| U3 - dose-specific alpha   | 14   | 3.3  | 1.5  | 0.5  | 0.1  | -     |
| Mean IPDE administrations  |      |      |      |      |      |       |
| U2 - shared alpha          | 4.2  | 1.6  | 0.5  | 0.2  | 0    | -     |
| U2 - dose-specific alpha   | 4.1  | 1.6  | 0.5  | 0.2  | 0    | -     |
| U3 - shared alpha          | 4.2  | 1.6  | 0.5  | 0.2  | 0    | -     |
| U3 - dose-specific alpha   | 4.2  | 1.5  | 0.5  | 0.2  | 0    | -     |
| Mean trial duration (days) |      |      |      |      |      | Days  |
| U2 - shared alpha          |      |      |      |      |      | 705.9 |
| U2 - dose-specific alpha   |      |      |      |      |      | 704.7 |
| U3 - shared alpha          |      |      |      |      |      | 704.7 |
| U3 - dose-specific alpha   |      |      |      |      |      | 705.8 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 9                 |        |        |        |        |        |              |
| DLT rate                   | 0.15   | 0.18   | 0.22   | 0.26   | 0.29   |              |
| Efficacy rate              | 0.05   | 0.08   | 0.20   | 0.25   | 0.50   |              |
| Utility 2                  | -0.1   | -0.1   | -0     | -0     | 0.2    |              |
| Utility 3                  | 37     | 37.6   | 43.2   | 44.6   | 58.4   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 25.4   | 11.5   | 16.2   | 18.7   | 27.6   | 0.6          |
| U2 - dose-specific alpha   | 25.5   | 11.3   | 16.5   | 18.7   | 27.4   | 0.6          |
| U3 - shared alpha          | 25.3   | 11.6   | 16.9   | 17.1   | 28.5   | 0.6          |
| U3 - dose-specific alpha   | 25.3   | 11.8   | 16.5   | 17.5   | 28.3   | 0.6          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 9.3    | 9.8    | 16.8   | 12.4   | 22.2   | 29.5         |
| U2 - dose-specific alpha   | 8.7    | 10.9   | 17.2   | 11.9   | 21.9   | 29.4         |
| U3 - shared alpha          | 13.6   | 10.8   | 15.5   | 10     | 21     | 29.1         |
| U3 - dose-specific alpha   | 14.3   | 10.2   | 15.3   | 10     | 20.6   | 29.6         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 7.3    | 4.3    | 5.3    | 3.7    | 4.5    | -            |
| U2 - dose-specific alpha   | 7.2    | 4.3    | 5.4    | 3.7    | 4.5    | -            |
| U3 - shared alpha          | 7.2    | 4.3    | 5.4    | 3.6    | 4.6    | -            |
| U3 - dose-specific alpha   | 7.3    | 4.2    | 5.4    | 3.7    | 4.5    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 1.9    | 2.5    | 1.8    | 1.5    | 1.3    | -            |
| U2 - dose-specific alpha   | 1.9    | 2.5    | 1.9    | 1.5    | 1.3    | -            |
| U3 - shared alpha          | 1.9    | 2.5    | 1.9    | 1.5    | 1.3    | -            |
| U3 - dose-specific alpha   | 1.9    | 2.4    | 1.9    | 1.5    | 1.3    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 882.4        |
| U2 - dose-specific alpha   |        |        |        |        |        | 882.2        |
| U3 - shared alpha          |        |        |        |        |        | 882.7        |
| U3 - dose-specific alpha   |        |        |        |        |        | 881.1        |

|                            |      |      |      |      |      |       |
|----------------------------|------|------|------|------|------|-------|
| Scenario 10                |      |      |      |      |      |       |
| DLT rate                   | 0.23 | 0.28 | 0.32 | 0.39 | 0.43 |       |
| Efficacy rate              | 0.05 | 0.40 | 0.42 | 0.45 | 0.50 |       |
| Utility 2                  | -0.2 | 0.1  | 0.1  | 0.1  | 0.1  |       |
| Utility 3                  | 33.8 | 52.8 | 52.4 | 51.4 | 52.8 |       |
| MTD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 53   | 18.2 | 14.6 | 7.5  | 2.7  | 4     |
| U2 - dose-specific alpha   | 52.7 | 18   | 15.3 | 7.3  | 2.6  | 4.1   |
| U3 - shared alpha          | 53.2 | 17.8 | 15   | 7.3  | 2.7  | 4     |
| U3 - dose-specific alpha   | 52.9 | 18   | 15.2 | 7.3  | 2.6  | 4     |
| OBD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 7.8  | 25.1 | 10.8 | 3.9  | 1.2  | 51.2  |
| U2 - dose-specific alpha   | 8    | 25.5 | 10.7 | 3.5  | 1.1  | 51.2  |
| U3 - shared alpha          | 8.1  | 26.4 | 10   | 2.9  | 1.1  | 51.5  |
| U3 - dose-specific alpha   | 8.5  | 27.7 | 9    | 2.9  | 0.9  | 51    |
| Mean administrations       |      |      |      |      |      |       |
| U2 - shared alpha          | 9.2  | 5.5  | 3.7  | 1.8  | 0.8  | -     |
| U2 - dose-specific alpha   | 9.2  | 5.5  | 3.7  | 1.8  | 0.7  | -     |
| U3 - shared alpha          | 9.2  | 5.5  | 3.7  | 1.8  | 0.7  | -     |
| U3 - dose-specific alpha   | 9.2  | 5.5  | 3.7  | 1.8  | 0.8  | -     |
| Mean IPDE administrations  |      |      |      |      |      |       |
| U2 - shared alpha          | 2.5  | 2.4  | 1    | 0.6  | 0.2  | -     |
| U2 - dose-specific alpha   | 2.5  | 2.5  | 1    | 0.6  | 0.2  | -     |
| U3 - shared alpha          | 2.5  | 2.4  | 1    | 0.6  | 0.2  | -     |
| U3 - dose-specific alpha   | 2.5  | 2.4  | 1    | 0.6  | 0.2  | -     |
| Mean trial duration (days) |      |      |      |      |      | Days  |
| U2 - shared alpha          |      |      |      |      |      | 776.2 |
| U2 - dose-specific alpha   |      |      |      |      |      | 776.6 |
| U3 - shared alpha          |      |      |      |      |      | 776.4 |
| U3 - dose-specific alpha   |      |      |      |      |      | 776   |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 11                |        |        |        |        |        |              |
| DLT rate                   | 0.22   | 0.32   | 0.44   | 0.59   | 0.67   |              |
| Efficacy rate              | 0.20   | 0.20   | 0.20   | 0.20   | 0.20   |              |
| Utility 2                  | -0     | -0.1   | -0.2   | -0.4   | -0.5   |              |
| Utility 3                  | 43.2   | 39.2   | 34.4   | 28.4   | 25.2   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 53.9   | 30.6   | 10.8   | 0.5    | 0      | 4.2          |
| U2 - dose-specific alpha   | 54.3   | 30.8   | 10.3   | 0.6    | 0      | 4            |
| U3 - shared alpha          | 54.9   | 29.2   | 11.3   | 0.4    | 0      | 4.2          |
| U3 - dose-specific alpha   | 54.1   | 30.8   | 10.6   | 0.3    | 0      | 4.2          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 63.8   | 18.9   | 2.6    | 0.2    | 0      | 14.5         |
| U2 - dose-specific alpha   | 63.9   | 18.9   | 2.7    | 0.1    | 0      | 14.4         |
| U3 - shared alpha          | 65.6   | 17.8   | 2      | 0.1    | 0      | 14.5         |
| U3 - dose-specific alpha   | 64.9   | 18.2   | 2.2    | 0.1    | 0      | 14.6         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 18.1   | 5.7    | 2.9    | 0.7    | 0.1    | -            |
| U2 - dose-specific alpha   | 18.1   | 5.7    | 2.8    | 0.7    | 0.1    | -            |
| U3 - shared alpha          | 18.1   | 5.7    | 2.9    | 0.7    | 0.1    | -            |
| U3 - dose-specific alpha   | 18.1   | 5.7    | 2.8    | 0.7    | 0.1    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 5.6    | 2.5    | 0.9    | 0.3    | 0      | -            |
| U2 - dose-specific alpha   | 5.6    | 2.5    | 0.9    | 0.3    | 0      | -            |
| U3 - shared alpha          | 5.6    | 2.5    | 0.9    | 0.3    | 0      | -            |
| U3 - dose-specific alpha   | 5.6    | 2.5    | 0.9    | 0.3    | 0      | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 998.9        |
| U2 - dose-specific alpha   |        |        |        |        |        | 998          |
| U3 - shared alpha          |        |        |        |        |        | 997.9        |
| U3 - dose-specific alpha   |        |        |        |        |        | 998.1        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 12                |      |      |      |      |      |        |
| DLT rate                   | 0.06 | 0.09 | 0.18 | 0.29 | 0.44 |        |
| Efficacy rate              | 0.05 | 0.10 | 0.12 | 0.40 | 0.60 |        |
| Utility 2                  | -0   | 0    | -0.1 | 0.1  | 0.2  |        |
| Utility 3                  | 40.6 | 42.4 | 40   | 52.4 | 58.4 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 4.4  | 2.7  | 24.7 | 45.2 | 22.9 | 0.1    |
| U2 - dose-specific alpha   | 4.4  | 2.7  | 25.2 | 44.4 | 23.2 | 0.1    |
| U3 - shared alpha          | 4.4  | 2.6  | 25.1 | 44.5 | 23.3 | 0.1    |
| U3 - dose-specific alpha   | 4.4  | 2.6  | 24.9 | 44.7 | 23.3 | 0.1    |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 13.8 | 13.9 | 11.2 | 38.9 | 16.8 | 5.4    |
| U2 - dose-specific alpha   | 14.1 | 14.8 | 11.5 | 36.5 | 17.4 | 5.7    |
| U3 - shared alpha          | 19.7 | 13.5 | 10.3 | 35.2 | 15.8 | 5.5    |
| U3 - dose-specific alpha   | 20.6 | 14.2 | 9.3  | 35.5 | 15.1 | 5.3    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 4.9  | 4.7  | 5.6  | 7.4  | 6.5  | -      |
| U2 - dose-specific alpha   | 4.9  | 4.8  | 5.6  | 7.2  | 6.5  | -      |
| U3 - shared alpha          | 4.9  | 4.8  | 5.6  | 7.3  | 6.5  | -      |
| U3 - dose-specific alpha   | 4.9  | 4.8  | 5.6  | 7.3  | 6.5  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.9  | 2.9  | 1.8  | 2.8  | 1.4  | -      |
| U2 - dose-specific alpha   | 0.9  | 3    | 1.8  | 2.7  | 1.5  | -      |
| U3 - shared alpha          | 0.9  | 3    | 1.8  | 2.7  | 1.5  | -      |
| U3 - dose-specific alpha   | 0.9  | 3    | 1.8  | 2.8  | 1.4  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1059.8 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1059.5 |
| U3 - shared alpha          |      |      |      |      |      | 1058.9 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1060.6 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 13                |        |        |        |        |        |              |
| DLT rate                   | 0.21   | 0.29   | 0.30   | 0.39   | 0.46   |              |
| Efficacy rate              | 0.05   | 0.30   | 0.32   | 0.35   | 0.40   |              |
| Utility 2                  | -0.2   | 0      | 0      | -0     | -0.1   |              |
| Utility 3                  | 34.6   | 46.4   | 47.2   | 45.4   | 45.6   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 48.8   | 20.3   | 17.6   | 8.1    | 2.4    | 2.8          |
| U2 - dose-specific alpha   | 48.8   | 20     | 17.7   | 8.4    | 2.3    | 2.8          |
| U3 - shared alpha          | 48.5   | 20.6   | 17.1   | 8.6    | 2.4    | 2.8          |
| U3 - dose-specific alpha   | 48.5   | 20.5   | 17.5   | 8.5    | 2.2    | 2.8          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 9.3    | 28.1   | 11.1   | 4      | 1.2    | 46.3         |
| U2 - dose-specific alpha   | 9.6    | 28     | 11.2   | 3.6    | 1.2    | 46.4         |
| U3 - shared alpha          | 11     | 28.8   | 9.9    | 3.1    | 1.1    | 46.1         |
| U3 - dose-specific alpha   | 10.4   | 29     | 9.6    | 3.8    | 1.1    | 46.1         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 9      | 6      | 4      | 1.9    | 0.8    | -            |
| U2 - dose-specific alpha   | 9      | 6      | 3.9    | 2      | 0.8    | -            |
| U3 - shared alpha          | 9      | 5.9    | 4      | 2      | 0.8    | -            |
| U3 - dose-specific alpha   | 9      | 6      | 4      | 2      | 0.8    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 2.4    | 2.8    | 1.2    | 0.7    | 0.2    | -            |
| U2 - dose-specific alpha   | 2.5    | 2.8    | 1.2    | 0.7    | 0.2    | -            |
| U3 - shared alpha          | 2.5    | 2.7    | 1.2    | 0.7    | 0.2    | -            |
| U3 - dose-specific alpha   | 2.5    | 2.8    | 1.2    | 0.7    | 0.2    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 788          |
| U2 - dose-specific alpha   |        |        |        |        |        | 789.8        |
| U3 - shared alpha          |        |        |        |        |        | 788.7        |
| U3 - dose-specific alpha   |        |        |        |        |        | 788          |

|                            |      |      |      |      |      |       |
|----------------------------|------|------|------|------|------|-------|
| Scenario 14                |      |      |      |      |      |       |
| DLT rate                   | 0.09 | 0.15 | 0.23 | 0.28 | 0.39 |       |
| Efficacy rate              | 0.05 | 0.08 | 0.12 | 0.30 | 0.40 |       |
| Utility 2                  | -0   | -0.1 | -0.1 | 0    | 0    |       |
| Utility 3                  | 39.4 | 38.8 | 38   | 46.8 | 48.4 |       |
| MTD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 10.9 | 8.9  | 26.2 | 29.8 | 24   | 0.2   |
| U2 - dose-specific alpha   | 11   | 8.8  | 26.1 | 29.9 | 24   | 0.2   |
| U3 - shared alpha          | 10.9 | 8.8  | 25.9 | 30.9 | 23.3 | 0.2   |
| U3 - dose-specific alpha   | 11.1 | 8.1  | 26.2 | 30.4 | 24   | 0.2   |
| OBD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 18.7 | 13.6 | 15.4 | 24.5 | 13.2 | 14.6  |
| U2 - dose-specific alpha   | 18   | 14.8 | 14.3 | 26.1 | 12.6 | 14.2  |
| U3 - shared alpha          | 25.2 | 15.6 | 11.1 | 23.9 | 9.7  | 14.5  |
| U3 - dose-specific alpha   | 25.4 | 15.1 | 11.3 | 24.5 | 9.4  | 14.3  |
| Mean administrations       |      |      |      |      |      |       |
| U2 - shared alpha          | 6.3  | 5.3  | 5.9  | 5.7  | 4.6  | -     |
| U2 - dose-specific alpha   | 6.2  | 5.4  | 5.8  | 5.7  | 4.6  | -     |
| U3 - shared alpha          | 6.2  | 5.3  | 6    | 5.7  | 4.6  | -     |
| U3 - dose-specific alpha   | 6.3  | 5.3  | 5.9  | 5.7  | 4.5  | -     |
| Mean IPDE administrations  |      |      |      |      |      |       |
| U2 - shared alpha          | 1.5  | 3.1  | 2    | 2.2  | 1.4  | -     |
| U2 - dose-specific alpha   | 1.4  | 3.1  | 2    | 2.2  | 1.4  | -     |
| U3 - shared alpha          | 1.4  | 3.1  | 2.1  | 2.2  | 1.4  | -     |
| U3 - dose-specific alpha   | 1.5  | 3.1  | 2.1  | 2.2  | 1.4  | -     |
| Mean trial duration (days) |      |      |      |      |      | Days  |
| U2 - shared alpha          |      |      |      |      |      | 967.2 |
| U2 - dose-specific alpha   |      |      |      |      |      | 968.8 |
| U3 - shared alpha          |      |      |      |      |      | 966.2 |
| U3 - dose-specific alpha   |      |      |      |      |      | 964.8 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 15                |        |        |        |        |        |              |
| DLT rate                   | 0.13   | 0.20   | 0.32   | 0.39   | 0.46   |              |
| Efficacy rate              | 0.10   | 0.10   | 0.10   | 0.10   | 0.10   |              |
| Utility 2                  | -0     | -0.1   | -0.2   | -0.3   | -0.4   |              |
| Utility 3                  | 40.8   | 38     | 33.2   | 30.4   | 27.6   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 20.6   | 23     | 34.6   | 16.9   | 4.5    | 0.4          |
| U2 - dose-specific alpha   | 20.7   | 23.3   | 34.6   | 16.6   | 4.4    | 0.4          |
| U3 - shared alpha          | 20.7   | 23.3   | 34.9   | 16.1   | 4.6    | 0.4          |
| U3 - dose-specific alpha   | 20.6   | 23     | 36     | 15.7   | 4.3    | 0.4          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 42.8   | 24.7   | 9.6    | 3.1    | 0.2    | 19.6         |
| U2 - dose-specific alpha   | 42.8   | 25.1   | 9.2    | 3      | 0.3    | 19.6         |
| U3 - shared alpha          | 50.2   | 21.9   | 6.5    | 1.5    | 0.3    | 19.6         |
| U3 - dose-specific alpha   | 50.7   | 22.8   | 5.7    | 1.4    | 0.1    | 19.3         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 10.6   | 7.1    | 5.7    | 2.9    | 1.1    | -            |
| U2 - dose-specific alpha   | 10.5   | 7      | 5.7    | 3      | 1.2    | -            |
| U3 - shared alpha          | 10.5   | 7      | 5.7    | 3      | 1.2    | -            |
| U3 - dose-specific alpha   | 10.6   | 6.9    | 5.7    | 2.9    | 1.2    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 3.2    | 3.6    | 2      | 1.3    | 0.5    | -            |
| U2 - dose-specific alpha   | 3.2    | 3.5    | 2      | 1.3    | 0.5    | -            |
| U3 - shared alpha          | 3.2    | 3.5    | 2      | 1.3    | 0.5    | -            |
| U3 - dose-specific alpha   | 3.2    | 3.5    | 2      | 1.3    | 0.5    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 929.1        |
| U2 - dose-specific alpha   |        |        |        |        |        | 928.5        |
| U3 - shared alpha          |        |        |        |        |        | 928.9        |
| U3 - dose-specific alpha   |        |        |        |        |        | 929.4        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 16                |      |      |      |      |      |        |
| DLT rate                   | 0.02 | 0.06 | 0.09 | 0.14 | 0.26 |        |
| Efficacy rate              | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 |        |
| Utility 2                  | 0.2  | 0.1  | 0.1  | 0.1  | -0.1 |        |
| Utility 3                  | 51.2 | 49.6 | 48.4 | 46.4 | 41.6 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0.7  | 0.6  | 2.5  | 10.5 | 85.7 | 0      |
| U2 - dose-specific alpha   | 0.7  | 0.6  | 2.8  | 10.2 | 85.7 | 0      |
| U3 - shared alpha          | 0.7  | 0.6  | 2.7  | 10.8 | 85.2 | 0      |
| U3 - dose-specific alpha   | 0.7  | 0.6  | 2.7  | 10.2 | 85.8 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 29   | 26   | 16.2 | 15.7 | 13   | 0.1    |
| U2 - dose-specific alpha   | 29.9 | 25   | 16.3 | 16.5 | 12.2 | 0.1    |
| U3 - shared alpha          | 38.9 | 28.3 | 13.5 | 13   | 6.2  | 0.1    |
| U3 - dose-specific alpha   | 36.7 | 29.2 | 13.2 | 13.9 | 6.9  | 0.1    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 5.8  | 5.7  | 5.5  | 5.8  | 7.2  | -      |
| U2 - dose-specific alpha   | 5.8  | 5.7  | 5.5  | 5.8  | 7.2  | -      |
| U3 - shared alpha          | 5.8  | 5.7  | 5.5  | 5.8  | 7.1  | -      |
| U3 - dose-specific alpha   | 5.8  | 5.8  | 5.5  | 5.8  | 7.2  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 1.2  | 3.2  | 1.9  | 2.6  | 2.7  | -      |
| U2 - dose-specific alpha   | 1.2  | 3.2  | 1.9  | 2.6  | 2.8  | -      |
| U3 - shared alpha          | 1.2  | 3.2  | 1.9  | 2.7  | 2.7  | -      |
| U3 - dose-specific alpha   | 1.2  | 3.2  | 1.9  | 2.6  | 2.7  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1007.4 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1008   |
| U3 - shared alpha          |      |      |      |      |      | 1008   |
| U3 - dose-specific alpha   |      |      |      |      |      | 1007.7 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 17                |        |        |        |        |        |              |
| DLT rate                   | 0.11   | 0.23   | 0.39   | 0.58   | 0.76   |              |
| Efficacy rate              | 0.30   | 0.50   | 0.50   | 0.50   | 0.50   |              |
| Utility 2                  | 0.2    | 0.3    | 0.1    | -0.1   | -0.3   |              |
| Utility 3                  | 53.6   | 60.8   | 54.4   | 46.8   | 39.6   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 18.2   | 42.7   | 34.1   | 4.8    | 0      | 0.2          |
| U2 - dose-specific alpha   | 18     | 41.4   | 35.7   | 4.7    | 0      | 0.2          |
| U3 - shared alpha          | 17.8   | 42.3   | 35.5   | 4.2    | 0      | 0.2          |
| U3 - dose-specific alpha   | 17.4   | 41.8   | 35.6   | 5      | 0      | 0.2          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 31.2   | 51.8   | 14.6   | 1.5    | 0      | 0.9          |
| U2 - dose-specific alpha   | 30.8   | 50.8   | 16.2   | 1.3    | 0      | 0.9          |
| U3 - shared alpha          | 33.3   | 51.9   | 12.8   | 1.1    | 0      | 0.9          |
| U3 - dose-specific alpha   | 33.3   | 52.2   | 12.6   | 1      | 0      | 0.9          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 10.6   | 10.1   | 6.8    | 2      | 0.3    | -            |
| U2 - dose-specific alpha   | 10.7   | 10     | 6.9    | 2      | 0.3    | -            |
| U3 - shared alpha          | 10.6   | 10.1   | 6.9    | 2      | 0.3    | -            |
| U3 - dose-specific alpha   | 10.7   | 10.1   | 6.9    | 1.9    | 0.3    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 2.7    | 3.5    | 1.5    | 0.6    | 0.1    | -            |
| U2 - dose-specific alpha   | 2.7    | 3.4    | 1.6    | 0.6    | 0.1    | -            |
| U3 - shared alpha          | 2.7    | 3.5    | 1.5    | 0.6    | 0.1    | -            |
| U3 - dose-specific alpha   | 2.7    | 3.5    | 1.6    | 0.5    | 0.1    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1187.6       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1186.5       |
| U3 - shared alpha          |        |        |        |        |        | 1187.2       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1187.1       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 18                |      |      |      |      |      |        |
| DLT rate                   | 0.02 | 0.04 | 0.10 | 0.20 | 0.35 |        |
| Efficacy rate              | 0.05 | 0.10 | 0.15 | 0.20 | 0.50 |        |
| Utility 2                  | 0    | 0.1  | 0.1  | 0    | 0.1  |        |
| Utility 3                  | 42.2 | 44.4 | 45   | 44   | 56   |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0.7  | 0.2  | 4.6  | 26   | 68.5 | 0      |
| U2 - dose-specific alpha   | 0.7  | 0.2  | 4.7  | 26.4 | 68   | 0      |
| U3 - shared alpha          | 0.7  | 0.3  | 4.5  | 27.6 | 66.9 | 0      |
| U3 - dose-specific alpha   | 0.7  | 0.2  | 4.7  | 26.6 | 67.8 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 9.1  | 9.2  | 10.8 | 14   | 55.8 | 1.1    |
| U2 - dose-specific alpha   | 10.4 | 10.1 | 9.6  | 14.4 | 54.6 | 0.9    |
| U3 - shared alpha          | 16.1 | 12.7 | 9.4  | 12.6 | 48.1 | 1.1    |
| U3 - dose-specific alpha   | 15.2 | 13.9 | 9.7  | 11.5 | 48.7 | 1      |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 3.7  | 3.9  | 4.7  | 5.6  | 12   | -      |
| U2 - dose-specific alpha   | 3.7  | 3.8  | 4.8  | 5.5  | 12   | -      |
| U3 - shared alpha          | 3.7  | 3.9  | 4.7  | 5.6  | 12   | -      |
| U3 - dose-specific alpha   | 3.7  | 3.9  | 4.7  | 5.5  | 12.1 | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.3  | 2.7  | 1.5  | 2.5  | 3.3  | -      |
| U2 - dose-specific alpha   | 0.3  | 2.7  | 1.6  | 2.5  | 3.4  | -      |
| U3 - shared alpha          | 0.3  | 2.7  | 1.6  | 2.5  | 3.3  | -      |
| U3 - dose-specific alpha   | 0.3  | 2.7  | 1.6  | 2.5  | 3.3  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1070.3 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1070.4 |
| U3 - shared alpha          |      |      |      |      |      | 1070.3 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1071.4 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 19                |        |        |        |        |        |              |
| DLT rate                   | 0.11   | 0.26   | 0.39   | 0.60   | 0.77   |              |
| Efficacy rate              | 0.08   | 0.30   | 0.32   | 0.35   | 0.40   |              |
| Utility 2                  | -0     | 0      | -0.1   | -0.2   | -0.4   |              |
| Utility 3                  | 40.4   | 47.6   | 43.6   | 37     | 33.2   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 25.6   | 43.7   | 27.5   | 3      | 0      | 0.2          |
| U2 - dose-specific alpha   | 25.9   | 43.5   | 27.7   | 2.7    | 0      | 0.2          |
| U3 - shared alpha          | 25.4   | 43.6   | 28.2   | 2.6    | 0      | 0.2          |
| U3 - dose-specific alpha   | 25.8   | 43.2   | 28.1   | 2.7    | 0      | 0.2          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 18.4   | 49.9   | 13.7   | 0.9    | 0      | 17.1         |
| U2 - dose-specific alpha   | 18.7   | 49.9   | 13.6   | 0.9    | 0      | 16.9         |
| U3 - shared alpha          | 20.3   | 50.7   | 11.2   | 0.6    | 0      | 17.2         |
| U3 - dose-specific alpha   | 20.6   | 49.8   | 11.4   | 0.7    | 0      | 17.5         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 8.2    | 10.5   | 6.5    | 1.7    | 0.3    | -            |
| U2 - dose-specific alpha   | 8.1    | 10.5   | 6.4    | 1.7    | 0.3    | -            |
| U3 - shared alpha          | 8.1    | 10.6   | 6.4    | 1.7    | 0.3    | -            |
| U3 - dose-specific alpha   | 8.1    | 10.5   | 6.5    | 1.7    | 0.3    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 2.1    | 4.5    | 1.8    | 0.6    | 0.1    | -            |
| U2 - dose-specific alpha   | 2.1    | 4.5    | 1.8    | 0.6    | 0.1    | -            |
| U3 - shared alpha          | 2.1    | 4.6    | 1.8    | 0.6    | 0.1    | -            |
| U3 - dose-specific alpha   | 2.1    | 4.5    | 1.9    | 0.6    | 0.1    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 986.3        |
| U2 - dose-specific alpha   |        |        |        |        |        | 986.3        |
| U3 - shared alpha          |        |        |        |        |        | 986.1        |
| U3 - dose-specific alpha   |        |        |        |        |        | 986.7        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 20                |      |      |      |      |      |        |
| DLT rate                   | 0.07 | 0.14 | 0.31 | 0.41 | 0.52 |        |
| Efficacy rate              | 0.05 | 0.25 | 0.25 | 0.25 | 0.25 |        |
| Utility 2                  | -0   | 0.1  | -0.1 | -0.2 | -0.3 |        |
| Utility 3                  | 40.2 | 49.4 | 42.6 | 38.6 | 34.2 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 7.2  | 15.2 | 47.1 | 25.7 | 4.7  | 0.1    |
| U2 - dose-specific alpha   | 7.1  | 14.7 | 48.6 | 25.1 | 4.4  | 0.1    |
| U3 - shared alpha          | 6.8  | 15.6 | 47   | 25.6 | 4.9  | 0.1    |
| U3 - dose-specific alpha   | 7.2  | 15.1 | 46.9 | 26   | 4.7  | 0.1    |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 9.8  | 45.8 | 28.8 | 8.1  | 0.7  | 6.8    |
| U2 - dose-specific alpha   | 11   | 44.4 | 27.7 | 9.2  | 0.8  | 6.9    |
| U3 - shared alpha          | 15.5 | 48.2 | 21.6 | 7.2  | 0.7  | 6.8    |
| U3 - dose-specific alpha   | 16   | 49   | 20.7 | 7.2  | 0.3  | 6.8    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 5.3  | 9    | 8.5  | 4.1  | 1.8  | -      |
| U2 - dose-specific alpha   | 5.3  | 9.1  | 8.4  | 4.1  | 1.8  | -      |
| U3 - shared alpha          | 5.3  | 9.1  | 8.3  | 4.1  | 1.7  | -      |
| U3 - dose-specific alpha   | 5.3  | 9    | 8.4  | 4.1  | 1.8  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 1    | 4.4  | 2.6  | 1.5  | 0.6  | -      |
| U2 - dose-specific alpha   | 1    | 4.4  | 2.6  | 1.5  | 0.6  | -      |
| U3 - shared alpha          | 1.1  | 4.4  | 2.6  | 1.6  | 0.6  | -      |
| U3 - dose-specific alpha   | 1    | 4.4  | 2.6  | 1.5  | 0.6  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1015.6 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1014.2 |
| U3 - shared alpha          |      |      |      |      |      | 1014.8 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1014.6 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 21                |        |        |        |        |        |              |
| DLT rate                   | 0.37   | 0.56   | 0.69   | 0.83   | 0.88   |              |
| Efficacy rate              | 0.50   | 0.50   | 0.50   | 0.50   | 0.50   |              |
| Utility 2                  | 0.1    | -0.1   | -0.2   | -0.3   | -0.4   |              |
| Utility 3                  | 55.2   | 47.6   | 42.4   | 36.8   | 34.8   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 56.1   | 1.9    | 0      | 0      | 0      | 42           |
| U2 - dose-specific alpha   | 56.3   | 1.7    | 0      | 0      | 0      | 42           |
| U3 - shared alpha          | 55.7   | 1.9    | 0      | 0      | 0      | 42.4         |
| U3 - dose-specific alpha   | 55     | 2.1    | 0      | 0      | 0      | 42.9         |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 54.9   | 0.7    | 0      | 0      | 0      | 44.4         |
| U2 - dose-specific alpha   | 54.4   | 1      | 0      | 0      | 0      | 44.6         |
| U3 - shared alpha          | 54.5   | 0.8    | 0      | 0      | 0      | 44.7         |
| U3 - dose-specific alpha   | 55     | 1      | 0      | 0      | 0      | 44           |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 20.4   | 1.9    | 0.4    | 0      | 0      | -            |
| U2 - dose-specific alpha   | 20.4   | 1.9    | 0.4    | 0      | 0      | -            |
| U3 - shared alpha          | 20.4   | 1.9    | 0.4    | 0      | 0      | -            |
| U3 - dose-specific alpha   | 20.4   | 1.9    | 0.4    | 0      | 0      | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 4.2    | 0.6    | 0.1    | 0      | 0      | -            |
| U2 - dose-specific alpha   | 4.1    | 0.6    | 0.1    | 0      | 0      | -            |
| U3 - shared alpha          | 4.1    | 0.6    | 0.1    | 0      | 0      | -            |
| U3 - dose-specific alpha   | 4.2    | 0.6    | 0.1    | 0      | 0      | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 987.5        |
| U2 - dose-specific alpha   |        |        |        |        |        | 988.1        |
| U3 - shared alpha          |        |        |        |        |        | 988          |
| U3 - dose-specific alpha   |        |        |        |        |        | 989.9        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 22                |      |      |      |      |      |        |
| DLT rate                   | 0.07 | 0.27 | 0.50 | 0.56 | 0.78 |        |
| Efficacy rate              | 0.08 | 0.25 | 0.25 | 0.50 | 0.50 |        |
| Utility 2                  | 0    | -0   | -0.2 | -0.1 | -0.3 |        |
| Utility 3                  | 42   | 44.2 | 35   | 47.6 | 38.8 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 20.2 | 52.9 | 25.2 | 1.6  | 0    | 0.1    |
| U2 - dose-specific alpha   | 20.2 | 52.4 | 25.2 | 2.1  | 0    | 0.1    |
| U3 - shared alpha          | 21.3 | 52   | 25.2 | 1.4  | 0    | 0.1    |
| U3 - dose-specific alpha   | 20.5 | 54   | 24.1 | 1.3  | 0    | 0.1    |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 23.3 | 54.3 | 10.5 | 1.3  | 0    | 10.6   |
| U2 - dose-specific alpha   | 24.4 | 54.3 | 9.5  | 1.3  | 0    | 10.5   |
| U3 - shared alpha          | 27.9 | 52.5 | 8.2  | 1.1  | 0    | 10.3   |
| U3 - dose-specific alpha   | 27.9 | 53   | 7.8  | 1    | 0    | 10.3   |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 8.1  | 12.6 | 5.9  | 1.3  | 0.2  | -      |
| U2 - dose-specific alpha   | 8.1  | 12.7 | 5.8  | 1.3  | 0.2  | -      |
| U3 - shared alpha          | 8    | 12.7 | 5.9  | 1.3  | 0.2  | -      |
| U3 - dose-specific alpha   | 8.1  | 12.6 | 5.9  | 1.3  | 0.2  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 2    | 5.4  | 1.8  | 0.5  | 0    | -      |
| U2 - dose-specific alpha   | 2    | 5.5  | 1.7  | 0.5  | 0    | -      |
| U3 - shared alpha          | 2    | 5.5  | 1.7  | 0.5  | 0    | -      |
| U3 - dose-specific alpha   | 2    | 5.4  | 1.7  | 0.5  | 0    | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1005.7 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1005.2 |
| U3 - shared alpha          |      |      |      |      |      | 1007.6 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1006.9 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 23                |        |        |        |        |        |              |
| DLT rate                   | 0.32   | 0.55   | 0.80   | 0.94   | 0.97   |              |
| Efficacy rate              | 0.05   | 0.05   | 0.05   | 0.05   | 0.05   |              |
| Utility 2                  | -0.3   | -0.5   | -0.8   | -0.9   | -0.9   |              |
| Utility 3                  | 30.2   | 21     | 11     | 5.4    | 4.2    |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 77.8   | 4.5    | 0      | 0      | 0      | 17.7         |
| U2 - dose-specific alpha   | 77.7   | 4.9    | 0      | 0      | 0      | 17.4         |
| U3 - shared alpha          | 77.9   | 4.6    | 0      | 0      | 0      | 17.5         |
| U3 - dose-specific alpha   | 78.2   | 4.6    | 0      | 0      | 0      | 17.2         |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 6.3    | 3.1    | 0      | 0      | 0      | 90.6         |
| U2 - dose-specific alpha   | 6.3    | 3.3    | 0      | 0      | 0      | 90.4         |
| U3 - shared alpha          | 6.3    | 3.1    | 0      | 0      | 0      | 90.6         |
| U3 - dose-specific alpha   | 6.3    | 3      | 0      | 0      | 0      | 90.7         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 11.7   | 2.1    | 0.4    | 0      | 0      | -            |
| U2 - dose-specific alpha   | 11.7   | 2.1    | 0.4    | 0      | 0      | -            |
| U3 - shared alpha          | 11.7   | 2.1    | 0.4    | 0      | 0      | -            |
| U3 - dose-specific alpha   | 11.7   | 2.1    | 0.4    | 0      | 0      | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 3.4    | 1.1    | 0.1    | 0      | 0      | -            |
| U2 - dose-specific alpha   | 3.4    | 1.1    | 0.1    | 0      | 0      | -            |
| U3 - shared alpha          | 3.3    | 1.1    | 0.1    | 0      | 0      | -            |
| U3 - dose-specific alpha   | 3.4    | 1.1    | 0.1    | 0      | 0      | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 517.6        |
| U2 - dose-specific alpha   |        |        |        |        |        | 517.4        |
| U3 - shared alpha          |        |        |        |        |        | 517.5        |
| U3 - dose-specific alpha   |        |        |        |        |        | 517.7        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 24                |      |      |      |      |      |        |
| DLT rate                   | 0.01 | 0.02 | 0.05 | 0.10 | 0.25 |        |
| Efficacy rate              | 0.05 | 0.08 | 0.20 | 0.50 | 0.50 |        |
| Utility 2                  | 0    | 0.1  | 0.1  | 0.4  | 0.2  |        |
| Utility 3                  | 42.6 | 44   | 50   | 66   | 60   |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0    | 0.1  | 0.6  | 4.4  | 94.9 | 0      |
| U2 - dose-specific alpha   | 0    | 0.1  | 0.6  | 4.2  | 95.1 | 0      |
| U3 - shared alpha          | 0    | 0.1  | 0.6  | 4.5  | 94.8 | 0      |
| U3 - dose-specific alpha   | 0    | 0.1  | 0.6  | 4.4  | 94.9 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0.5  | 1.7  | 4.4  | 46.4 | 46.9 | 0.1    |
| U2 - dose-specific alpha   | 1.1  | 1.9  | 4.3  | 46.8 | 45.7 | 0.2    |
| U3 - shared alpha          | 2.1  | 2.6  | 6    | 48.2 | 41   | 0.1    |
| U3 - dose-specific alpha   | 2.3  | 2.7  | 7.1  | 47   | 40.8 | 0.1    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 3.2  | 3.2  | 3.7  | 8.3  | 11.5 | -      |
| U2 - dose-specific alpha   | 3.2  | 3.2  | 3.7  | 8.4  | 11.5 | -      |
| U3 - shared alpha          | 3.2  | 3.2  | 3.7  | 8.4  | 11.5 | -      |
| U3 - dose-specific alpha   | 3.2  | 3.2  | 3.7  | 8.3  | 11.6 | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.1  | 2.4  | 1.2  | 3.2  | 3.2  | -      |
| U2 - dose-specific alpha   | 0.1  | 2.4  | 1.2  | 3.2  | 3.2  | -      |
| U3 - shared alpha          | 0.1  | 2.4  | 1.2  | 3.3  | 3.1  | -      |
| U3 - dose-specific alpha   | 0.1  | 2.4  | 1.2  | 3.2  | 3.2  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1097.9 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1099.1 |
| U3 - shared alpha          |      |      |      |      |      | 1098.5 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1099.3 |

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 25                |        |        |        |        |        |              |
| DLT rate                   | 0.09   | 0.18   | 0.37   | 0.63   | 0.77   |              |
| Efficacy rate              | 0.20   | 0.25   | 0.40   | 0.45   | 0.50   |              |
| Utility 2                  | 0.1    | 0.1    | 0      | -0.2   | -0.3   |              |
| Utility 3                  | 48.4   | 47.8   | 49.2   | 41.8   | 39.2   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 12.7   | 35     | 45     | 7.1    | 0      | 0.2          |
| U2 - dose-specific alpha   | 12.7   | 34.8   | 45.1   | 7.1    | 0.1    | 0.2          |
| U3 - shared alpha          | 12.4   | 34.9   | 45.4   | 7.1    | 0      | 0.2          |
| U3 - dose-specific alpha   | 12.4   | 35     | 45.3   | 7.1    | 0      | 0.2          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 33.3   | 32.2   | 28.4   | 4.1    | 0      | 2            |
| U2 - dose-specific alpha   | 33.6   | 31.4   | 28.9   | 4.2    | 0      | 1.9          |
| U3 - shared alpha          | 39.7   | 31.5   | 23.8   | 3.1    | 0      | 1.9          |
| U3 - dose-specific alpha   | 40.9   | 31.2   | 23.1   | 2.9    | 0      | 1.9          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 9.4    | 8.5    | 9      | 2.3    | 0.4    | -            |
| U2 - dose-specific alpha   | 9.5    | 8.4    | 9      | 2.3    | 0.4    | -            |
| U3 - shared alpha          | 9.5    | 8.5    | 9      | 2.3    | 0.4    | -            |
| U3 - dose-specific alpha   | 9.4    | 8.5    | 9      | 2.4    | 0.4    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 2.6    | 3.7    | 2.5    | 0.8    | 0.1    | -            |
| U2 - dose-specific alpha   | 2.6    | 3.7    | 2.5    | 0.8    | 0.1    | -            |
| U3 - shared alpha          | 2.6    | 3.7    | 2.5    | 0.8    | 0.1    | -            |
| U3 - dose-specific alpha   | 2.6    | 3.7    | 2.5    | 0.8    | 0.1    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1105.9       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1104.9       |
| U3 - shared alpha          |        |        |        |        |        | 1105.5       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1105.3       |

|                            |      |      |      |      |      |       |
|----------------------------|------|------|------|------|------|-------|
| Scenario 26                |      |      |      |      |      |       |
| DLT rate                   | 0.15 | 0.22 | 0.30 | 0.62 | 0.80 |       |
| Efficacy rate              | 0.10 | 0.20 | 0.35 | 0.45 | 0.55 |       |
| Utility 2                  | -0.1 | -0   | 0.1  | -0.2 | -0.2 |       |
| Utility 3                  | 40   | 43.2 | 49   | 42.2 | 41   |       |
| MTD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 28.4 | 30.2 | 33.6 | 7.1  | 0.1  | 0.6   |
| U2 - dose-specific alpha   | 28.7 | 30.4 | 32.3 | 7.9  | 0.1  | 0.6   |
| U3 - shared alpha          | 28.4 | 30.4 | 32.8 | 7.8  | 0    | 0.6   |
| U3 - dose-specific alpha   | 28.9 | 29.5 | 33.5 | 7.5  | 0    | 0.6   |
| OBD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 22.2 | 29.6 | 25.2 | 3.1  | 0    | 19.9  |
| U2 - dose-specific alpha   | 22   | 30   | 24.4 | 3.8  | 0    | 19.8  |
| U3 - shared alpha          | 26   | 28   | 21.8 | 3.8  | 0    | 20.4  |
| U3 - dose-specific alpha   | 26.7 | 26.9 | 23   | 3.1  | 0    | 20.3  |
| Mean administrations       |      |      |      |      |      |       |
| U2 - shared alpha          | 9.9  | 6.9  | 7.5  | 2.3  | 0.3  | -     |
| U2 - dose-specific alpha   | 9.8  | 6.9  | 7.6  | 2.3  | 0.3  | -     |
| U3 - shared alpha          | 9.9  | 6.8  | 7.6  | 2.3  | 0.3  | -     |
| U3 - dose-specific alpha   | 9.9  | 7    | 7.5  | 2.3  | 0.3  | -     |
| Mean IPDE administrations  |      |      |      |      |      |       |
| U2 - shared alpha          | 2.9  | 3.3  | 2.2  | 0.8  | 0.1  | -     |
| U2 - dose-specific alpha   | 2.8  | 3.3  | 2.2  | 0.8  | 0.1  | -     |
| U3 - shared alpha          | 2.9  | 3.3  | 2.2  | 0.8  | 0.1  | -     |
| U3 - dose-specific alpha   | 2.9  | 3.3  | 2.2  | 0.8  | 0.1  | -     |
| Mean trial duration (days) |      |      |      |      |      | Days  |
| U2 - shared alpha          |      |      |      |      |      | 973.6 |
| U2 - dose-specific alpha   |      |      |      |      |      | 973.4 |
| U3 - shared alpha          |      |      |      |      |      | 971.7 |
| U3 - dose-specific alpha   |      |      |      |      |      | 971.9 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 27                |        |        |        |        |        |              |
| DLT rate                   | 0.11   | 0.17   | 0.30   | 0.50   | 0.62   |              |
| Efficacy rate              | 0.10   | 0.20   | 0.40   | 0.42   | 0.45   |              |
| Utility 2                  | -0     | 0      | 0.1    | -0.1   | -0.2   |              |
| Utility 3                  | 41.6   | 45.2   | 52     | 45.2   | 42.2   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 15.8   | 25     | 43.2   | 15     | 0.8    | 0.2          |
| U2 - dose-specific alpha   | 16.2   | 24.3   | 43.6   | 15     | 0.7    | 0.2          |
| U3 - shared alpha          | 16     | 24.7   | 44.1   | 14.2   | 0.8    | 0.2          |
| U3 - dose-specific alpha   | 16.1   | 25.1   | 43.6   | 14.3   | 0.7    | 0.2          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 18     | 25.1   | 37.4   | 6.9    | 0.3    | 12.3         |
| U2 - dose-specific alpha   | 17.4   | 24.6   | 39     | 6.3    | 0.3    | 12.4         |
| U3 - shared alpha          | 20.8   | 25.2   | 36.4   | 5.4    | 0.1    | 12.1         |
| U3 - dose-specific alpha   | 21.6   | 25.9   | 34.4   | 5.8    | 0.1    | 12.2         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 7.6    | 6.7    | 9.1    | 3.6    | 1.1    | -            |
| U2 - dose-specific alpha   | 7.7    | 6.7    | 9.1    | 3.5    | 1.1    | -            |
| U3 - shared alpha          | 7.6    | 6.7    | 9.1    | 3.6    | 1.1    | -            |
| U3 - dose-specific alpha   | 7.7    | 6.7    | 9      | 3.6    | 1.1    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 2      | 3.3    | 2.6    | 1.1    | 0.3    | -            |
| U2 - dose-specific alpha   | 2      | 3.3    | 2.6    | 1.1    | 0.3    | -            |
| U3 - shared alpha          | 2      | 3.3    | 2.6    | 1.1    | 0.3    | -            |
| U3 - dose-specific alpha   | 2      | 3.3    | 2.6    | 1.1    | 0.3    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1037.9       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1036.7       |
| U3 - shared alpha          |        |        |        |        |        | 1037.9       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1037         |

|                            |      |      |      |      |      |       |
|----------------------------|------|------|------|------|------|-------|
| Scenario 28                |      |      |      |      |      |       |
| DLT rate                   | 0.20 | 0.30 | 0.50 | 0.60 | 0.77 |       |
| Efficacy rate              | 0.05 | 0.30 | 0.32 | 0.35 | 0.40 |       |
| Utility 2                  | -0.1 | 0    | -0.2 | -0.2 | -0.4 |       |
| Utility 3                  | 35   | 46   | 39.2 | 37   | 33.2 |       |
| MTD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 57.4 | 30.4 | 9.5  | 0.3  | 0    | 2.4   |
| U2 - dose-specific alpha   | 56.8 | 31.1 | 9.4  | 0.1  | 0    | 2.6   |
| U3 - shared alpha          | 57   | 30.9 | 9.4  | 0.1  | 0    | 2.6   |
| U3 - dose-specific alpha   | 57.7 | 30.5 | 8.8  | 0.4  | 0    | 2.6   |
| OBD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 16.7 | 31.1 | 4.8  | 0.1  | 0    | 47.3  |
| U2 - dose-specific alpha   | 16.6 | 30.9 | 5.1  | 0    | 0    | 47.4  |
| U3 - shared alpha          | 17.3 | 31.9 | 3.6  | 0    | 0    | 47.2  |
| U3 - dose-specific alpha   | 17.5 | 30.7 | 3.7  | 0.1  | 0    | 48    |
| Mean administrations       |      |      |      |      |      |       |
| U2 - shared alpha          | 9.4  | 8.1  | 3.5  | 0.6  | 0.1  | -     |
| U2 - dose-specific alpha   | 9.5  | 8.1  | 3.5  | 0.6  | 0.1  | -     |
| U3 - shared alpha          | 9.4  | 8.2  | 3.5  | 0.6  | 0.1  | -     |
| U3 - dose-specific alpha   | 9.4  | 8.1  | 3.5  | 0.6  | 0.1  | -     |
| Mean IPDE administrations  |      |      |      |      |      |       |
| U2 - shared alpha          | 2.6  | 3.4  | 1    | 0.2  | 0    | -     |
| U2 - dose-specific alpha   | 2.6  | 3.4  | 1    | 0.2  | 0    | -     |
| U3 - shared alpha          | 2.6  | 3.4  | 1    | 0.2  | 0    | -     |
| U3 - dose-specific alpha   | 2.6  | 3.4  | 1    | 0.2  | 0    | -     |
| Mean trial duration (days) |      |      |      |      |      | Days  |
| U2 - shared alpha          |      |      |      |      |      | 797.4 |
| U2 - dose-specific alpha   |      |      |      |      |      | 797.9 |
| U3 - shared alpha          |      |      |      |      |      | 798.2 |
| U3 - dose-specific alpha   |      |      |      |      |      | 796.9 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 29                |        |        |        |        |        |              |
| DLT rate                   | 0.19   | 0.30   | 0.56   | 0.74   | 0.86   |              |
| Efficacy rate              | 0.40   | 0.40   | 0.40   | 0.40   | 0.40   |              |
| Utility 2                  | 0.2    | 0.1    | -0.2   | -0.3   | -0.5   |              |
| Utility 3                  | 56.4   | 52     | 41.6   | 34.4   | 29.6   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 46.8   | 39.1   | 11     | 0.4    | 0      | 2.7          |
| U2 - dose-specific alpha   | 47.1   | 38.5   | 11.2   | 0.5    | 0      | 2.7          |
| U3 - shared alpha          | 47     | 38.9   | 10.8   | 0.6    | 0      | 2.7          |
| U3 - dose-specific alpha   | 46.7   | 39.2   | 10.9   | 0.5    | 0      | 2.7          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 71     | 23     | 3      | 0.1    | 0      | 2.9          |
| U2 - dose-specific alpha   | 70.8   | 23.1   | 2.9    | 0.2    | 0      | 3            |
| U3 - shared alpha          | 72.8   | 21.7   | 2.3    | 0.2    | 0      | 3            |
| U3 - dose-specific alpha   | 73.1   | 21.4   | 2.4    | 0.1    | 0      | 3            |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 18.6   | 7.2    | 3.1    | 0.4    | 0      | -            |
| U2 - dose-specific alpha   | 18.7   | 7.2    | 3.1    | 0.4    | 0      | -            |
| U3 - shared alpha          | 18.7   | 7.2    | 3.1    | 0.4    | 0      | -            |
| U3 - dose-specific alpha   | 18.6   | 7.3    | 3.1    | 0.4    | 0      | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 4.8    | 2.4    | 0.9    | 0.1    | 0      | -            |
| U2 - dose-specific alpha   | 4.9    | 2.4    | 0.9    | 0.1    | 0      | -            |
| U3 - shared alpha          | 4.8    | 2.4    | 0.8    | 0.1    | 0      | -            |
| U3 - dose-specific alpha   | 4.8    | 2.4    | 0.8    | 0.1    | 0      | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1170.1       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1169.8       |
| U3 - shared alpha          |        |        |        |        |        | 1170.3       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1169.9       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 30                |      |      |      |      |      |        |
| DLT rate                   | 0.05 | 0.08 | 0.17 | 0.27 | 0.54 |        |
| Efficacy rate              | 0.12 | 0.15 | 0.20 | 0.40 | 0.50 |        |
| Utility 2                  | 0.1  | 0.1  | 0    | 0.1  | -0   |        |
| Utility 3                  | 45.2 | 45.8 | 45.2 | 53.2 | 48.4 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 2.6  | 2.3  | 26.2 | 47.7 | 21.2 | 0      |
| U2 - dose-specific alpha   | 2.6  | 2.4  | 25.8 | 48.1 | 21.1 | 0      |
| U3 - shared alpha          | 2.6  | 2.4  | 25.7 | 48   | 21.3 | 0      |
| U3 - dose-specific alpha   | 2.6  | 2    | 26.9 | 47.6 | 20.9 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 18.8 | 15.5 | 15.6 | 37.6 | 10.9 | 1.6    |
| U2 - dose-specific alpha   | 18.7 | 15.7 | 17.5 | 36.2 | 10.3 | 1.6    |
| U3 - shared alpha          | 26.2 | 17.5 | 13.8 | 32.5 | 8.4  | 1.6    |
| U3 - dose-specific alpha   | 25.5 | 17.6 | 13.8 | 32.2 | 9.2  | 1.7    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 5.2  | 4.6  | 6.2  | 8.1  | 5.6  | -      |
| U2 - dose-specific alpha   | 5.1  | 4.7  | 6.2  | 8    | 5.7  | -      |
| U3 - shared alpha          | 5.2  | 4.6  | 6.2  | 8.1  | 5.7  | -      |
| U3 - dose-specific alpha   | 5.2  | 4.7  | 6.1  | 8.1  | 5.6  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 1    | 2.8  | 2    | 2.8  | 1.4  | -      |
| U2 - dose-specific alpha   | 1    | 2.8  | 2    | 2.8  | 1.4  | -      |
| U3 - shared alpha          | 1    | 2.8  | 2    | 2.8  | 1.4  | -      |
| U3 - dose-specific alpha   | 1    | 2.8  | 1.9  | 2.8  | 1.4  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1089.8 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1090   |
| U3 - shared alpha          |      |      |      |      |      | 1090.9 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1089.3 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 31                |        |        |        |        |        |              |
| DLT rate                   | 0.04   | 0.06   | 0.10   | 0.20   | 0.60   |              |
| Efficacy rate              | 0.20   | 0.50   | 0.50   | 0.50   | 0.50   |              |
| Utility 2                  | 0.2    | 0.4    | 0.4    | 0.3    | -0.1   |              |
| Utility 3                  | 50.4   | 67.6   | 66     | 62     | 46     |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 1.6    | 1.3    | 10.1   | 43.6   | 43.4   | 0            |
| U2 - dose-specific alpha   | 1.6    | 1.4    | 9.4    | 43.5   | 44.1   | 0            |
| U3 - shared alpha          | 1.6    | 1.3    | 9.9    | 41.9   | 45.3   | 0            |
| U3 - dose-specific alpha   | 1.6    | 1.4    | 9.5    | 43.2   | 44.3   | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 6      | 34.7   | 27     | 24.5   | 7.5    | 0.3          |
| U2 - dose-specific alpha   | 5.9    | 35.8   | 27.1   | 23.5   | 7.4    | 0.3          |
| U3 - shared alpha          | 8.9    | 39.9   | 26     | 21     | 3.9    | 0.3          |
| U3 - dose-specific alpha   | 8.5    | 43     | 24.9   | 18.9   | 4.4    | 0.3          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 4.1    | 5.7    | 6.4    | 7.9    | 5.8    | -            |
| U2 - dose-specific alpha   | 4.1    | 5.8    | 6.4    | 7.9    | 5.7    | -            |
| U3 - shared alpha          | 4.1    | 5.7    | 6.4    | 7.9    | 5.7    | -            |
| U3 - dose-specific alpha   | 4.1    | 5.7    | 6.5    | 7.9    | 5.8    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 0.5    | 2.8    | 1.6    | 2      | 1.5    | -            |
| U2 - dose-specific alpha   | 0.5    | 2.9    | 1.6    | 2      | 1.5    | -            |
| U3 - shared alpha          | 0.5    | 2.8    | 1.7    | 2      | 1.5    | -            |
| U3 - dose-specific alpha   | 0.5    | 2.9    | 1.6    | 2      | 1.5    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1185.1       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1185.1       |
| U3 - shared alpha          |        |        |        |        |        | 1184.3       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1185.5       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 32                |      |      |      |      |      |        |
| DLT rate                   | 0.02 | 0.05 | 0.11 | 0.30 | 0.36 |        |
| Efficacy rate              | 0.08 | 0.15 | 0.30 | 0.30 | 0.30 |        |
| Utility 2                  | 0.1  | 0.1  | 0.2  | 0    | -0.1 |        |
| Utility 3                  | 44   | 47   | 53.6 | 46   | 43.6 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0.7  | 0.7  | 7.6  | 37.4 | 53.6 | 0      |
| U2 - dose-specific alpha   | 0.7  | 0.5  | 7.9  | 37.5 | 53.4 | 0      |
| U3 - shared alpha          | 0.7  | 0.5  | 8.1  | 36.5 | 54.2 | 0      |
| U3 - dose-specific alpha   | 0.7  | 0.5  | 7.7  | 37.4 | 53.7 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 8.4  | 12.8 | 37.1 | 28.4 | 12.5 | 0.8    |
| U2 - dose-specific alpha   | 8.4  | 14.5 | 34.9 | 28.2 | 13.2 | 0.8    |
| U3 - shared alpha          | 13.5 | 18.5 | 36.1 | 25   | 6.1  | 0.8    |
| U3 - dose-specific alpha   | 13.5 | 19.4 | 33.7 | 25.8 | 6.8  | 0.8    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 3.9  | 4.3  | 7.8  | 7.3  | 6.5  | -      |
| U2 - dose-specific alpha   | 4    | 4.3  | 7.7  | 7.3  | 6.6  | -      |
| U3 - shared alpha          | 4    | 4.3  | 7.8  | 7.3  | 6.5  | -      |
| U3 - dose-specific alpha   | 4    | 4.3  | 7.7  | 7.3  | 6.5  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.4  | 2.8  | 2.5  | 2.8  | 2.2  | -      |
| U2 - dose-specific alpha   | 0.4  | 2.9  | 2.5  | 2.7  | 2.2  | -      |
| U3 - shared alpha          | 0.4  | 2.8  | 2.5  | 2.7  | 2.2  | -      |
| U3 - dose-specific alpha   | 0.4  | 2.8  | 2.5  | 2.7  | 2.2  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1056.2 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1056.7 |
| U3 - shared alpha          |      |      |      |      |      | 1054.6 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1054.9 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 33                |        |        |        |        |        |              |
| DLT rate                   | 0.01   | 0.02   | 0.09   | 0.32   | 0.38   |              |
| Efficacy rate              | 0.05   | 0.08   | 0.20   | 0.40   | 0.50   |              |
| Utility 2                  | 0      | 0.1    | 0.1    | 0.1    | 0.1    |              |
| Utility 3                  | 42.6   | 44     | 48.4   | 51.2   | 54.8   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 0      | 0.8    | 9.6    | 43.5   | 46.1   | 0            |
| U2 - dose-specific alpha   | 0      | 0.6    | 9.1    | 43.4   | 46.9   | 0            |
| U3 - shared alpha          | 0      | 0.7    | 9.4    | 43.6   | 46.3   | 0            |
| U3 - dose-specific alpha   | 0      | 0.4    | 10.1   | 42.7   | 46.8   | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 7.9    | 7.2    | 13.7   | 41.9   | 29.2   | 0.1          |
| U2 - dose-specific alpha   | 7.9    | 6.8    | 13.3   | 42     | 29.8   | 0.2          |
| U3 - shared alpha          | 12.4   | 8.9    | 14     | 39.8   | 24.8   | 0.1          |
| U3 - dose-specific alpha   | 13.5   | 9      | 12.7   | 39.3   | 25.4   | 0.1          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 3.5    | 3.7    | 5.3    | 8.2    | 9.4    | -            |
| U2 - dose-specific alpha   | 3.5    | 3.6    | 5.3    | 8.1    | 9.5    | -            |
| U3 - shared alpha          | 3.5    | 3.6    | 5.2    | 8.2    | 9.4    | -            |
| U3 - dose-specific alpha   | 3.5    | 3.6    | 5.3    | 8.2    | 9.5    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 0.2    | 2.7    | 1.6    | 3.1    | 2.4    | -            |
| U2 - dose-specific alpha   | 0.2    | 2.6    | 1.6    | 3.1    | 2.4    | -            |
| U3 - shared alpha          | 0.2    | 2.6    | 1.6    | 3.1    | 2.3    | -            |
| U3 - dose-specific alpha   | 0.2    | 2.6    | 1.6    | 3.1    | 2.4    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1105.5       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1105.4       |
| U3 - shared alpha          |        |        |        |        |        | 1106.4       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1106         |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 34                |      |      |      |      |      |        |
| DLT rate                   | 0.05 | 0.13 | 0.30 | 0.36 | 0.42 |        |
| Efficacy rate              | 0.05 | 0.10 | 0.30 | 0.32 | 0.35 |        |
| Utility 2                  | 0    | -0   | 0    | -0   | -0.1 |        |
| Utility 3                  | 41   | 40.8 | 46   | 44.8 | 44.2 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 3.2  | 11.7 | 46.8 | 26.5 | 11.8 | 0      |
| U2 - dose-specific alpha   | 3.3  | 11.6 | 46.7 | 26.5 | 11.9 | 0      |
| U3 - shared alpha          | 3.3  | 11.4 | 47.1 | 26.7 | 11.5 | 0      |
| U3 - dose-specific alpha   | 3.2  | 10.5 | 47.2 | 27.6 | 11.5 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 13.8 | 12.8 | 49.5 | 15.6 | 3.4  | 4.9    |
| U2 - dose-specific alpha   | 13.3 | 13   | 48.7 | 15.6 | 4.2  | 5.2    |
| U3 - shared alpha          | 18.6 | 13.5 | 46.9 | 13.5 | 2.2  | 5.3    |
| U3 - dose-specific alpha   | 17.7 | 14.5 | 46.3 | 14.5 | 2.2  | 4.8    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 5.1  | 5.9  | 10   | 5.3  | 2.8  | -      |
| U2 - dose-specific alpha   | 5.2  | 5.9  | 9.9  | 5.3  | 2.8  | -      |
| U3 - shared alpha          | 5.2  | 5.9  | 10   | 5.2  | 2.8  | -      |
| U3 - dose-specific alpha   | 5.1  | 5.8  | 10   | 5.3  | 2.9  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 1    | 3.3  | 3.2  | 1.8  | 0.9  | -      |
| U2 - dose-specific alpha   | 1    | 3.3  | 3.1  | 1.9  | 0.9  | -      |
| U3 - shared alpha          | 1    | 3.3  | 3.2  | 1.8  | 0.9  | -      |
| U3 - dose-specific alpha   | 1    | 3.3  | 3.2  | 1.9  | 0.9  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1039.1 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1039.3 |
| U3 - shared alpha          |      |      |      |      |      | 1039.3 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1040.2 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 35                |        |        |        |        |        |              |
| DLT rate                   | 0.03   | 0.09   | 0.30   | 0.40   | 0.45   |              |
| Efficacy rate              | 0.20   | 0.45   | 0.45   | 0.45   | 0.45   |              |
| Utility 2                  | 0.2    | 0.4    | 0.1    | 0.1    | 0      |              |
| Utility 3                  | 50.8   | 63.4   | 55     | 51     | 49     |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 1.6    | 8.3    | 46.7   | 31     | 12.4   | 0            |
| U2 - dose-specific alpha   | 1.5    | 7.7    | 46.8   | 30.6   | 13.4   | 0            |
| U3 - shared alpha          | 1.6    | 8.2    | 48.2   | 28.9   | 13.1   | 0            |
| U3 - dose-specific alpha   | 1.6    | 7.3    | 48.1   | 30.2   | 12.8   | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 11     | 42.5   | 33     | 10.8   | 2.5    | 0.2          |
| U2 - dose-specific alpha   | 11.8   | 40.7   | 34     | 10.4   | 2.9    | 0.2          |
| U3 - shared alpha          | 15.8   | 47.4   | 26.1   | 8.8    | 1.7    | 0.2          |
| U3 - dose-specific alpha   | 15.6   | 47.5   | 27.1   | 8      | 1.6    | 0.2          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 4.5    | 8.8    | 9      | 5.1    | 2.7    | -            |
| U2 - dose-specific alpha   | 4.6    | 8.6    | 9.1    | 5      | 2.7    | -            |
| U3 - shared alpha          | 4.5    | 8.7    | 9      | 5.1    | 2.6    | -            |
| U3 - dose-specific alpha   | 4.5    | 8.6    | 9      | 5.1    | 2.7    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 0.6    | 3.8    | 2.2    | 1.5    | 0.7    | -            |
| U2 - dose-specific alpha   | 0.6    | 3.7    | 2.3    | 1.5    | 0.7    | -            |
| U3 - shared alpha          | 0.6    | 3.8    | 2.3    | 1.5    | 0.7    | -            |
| U3 - dose-specific alpha   | 0.6    | 3.7    | 2.2    | 1.5    | 0.7    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1165.9       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1165.7       |
| U3 - shared alpha          |        |        |        |        |        | 1165.3       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1165         |

|                            |      |      |      |      |      |       |
|----------------------------|------|------|------|------|------|-------|
| Scenario 36                |      |      |      |      |      |       |
| DLT rate                   | 0.13 | 0.29 | 0.35 | 0.40 | 0.43 |       |
| Efficacy rate              | 0.05 | 0.20 | 0.22 | 0.30 | 0.35 |       |
| Utility 2                  | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 |       |
| Utility 3                  | 37.8 | 40.4 | 39.2 | 42   | 43.8 |       |
| MTD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 31.4 | 34   | 21.6 | 10.1 | 2.5  | 0.4   |
| U2 - dose-specific alpha   | 31   | 35.2 | 21.4 | 9.3  | 2.7  | 0.4   |
| U3 - shared alpha          | 31.5 | 34.3 | 21.3 | 10.1 | 2.4  | 0.4   |
| U3 - dose-specific alpha   | 31.1 | 34.3 | 21.9 | 9.5  | 2.8  | 0.4   |
| OBD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 16.5 | 37.1 | 12.9 | 5.3  | 0.8  | 27.4  |
| U2 - dose-specific alpha   | 17.2 | 37.4 | 12.9 | 4.5  | 1    | 27    |
| U3 - shared alpha          | 20.6 | 36.2 | 10.6 | 4.3  | 0.7  | 27.6  |
| U3 - dose-specific alpha   | 20.2 | 37.1 | 10.7 | 4.1  | 0.8  | 27.1  |
| Mean administrations       |      |      |      |      |      |       |
| U2 - shared alpha          | 8.3  | 8.4  | 5.2  | 2.3  | 1    | -     |
| U2 - dose-specific alpha   | 8.2  | 8.5  | 5.2  | 2.3  | 0.9  | -     |
| U3 - shared alpha          | 8.3  | 8.4  | 5.2  | 2.4  | 0.9  | -     |
| U3 - dose-specific alpha   | 8.3  | 8.3  | 5.3  | 2.4  | 1    | -     |
| Mean IPDE administrations  |      |      |      |      |      |       |
| U2 - shared alpha          | 2.2  | 4    | 1.7  | 0.9  | 0.3  | -     |
| U2 - dose-specific alpha   | 2.1  | 4    | 1.7  | 0.9  | 0.3  | -     |
| U3 - shared alpha          | 2.2  | 4    | 1.7  | 0.9  | 0.3  | -     |
| U3 - dose-specific alpha   | 2.2  | 3.9  | 1.7  | 0.9  | 0.3  | -     |
| Mean trial duration (days) |      |      |      |      |      | Days  |
| U2 - shared alpha          |      |      |      |      |      | 890.1 |
| U2 - dose-specific alpha   |      |      |      |      |      | 890.1 |
| U3 - shared alpha          |      |      |      |      |      | 892.1 |
| U3 - dose-specific alpha   |      |      |      |      |      | 890.9 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 37                |        |        |        |        |        |              |
| DLT rate                   | 0.08   | 0.30   | 0.40   | 0.45   | 0.50   |              |
| Efficacy rate              | 0.05   | 0.20   | 0.20   | 0.20   | 0.20   |              |
| Utility 2                  | -0     | -0.1   | -0.2   | -0.2   | -0.3   |              |
| Utility 3                  | 39.8   | 40     | 36     | 34     | 32     |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 23.5   | 44.8   | 24.5   | 6.3    | 0.8    | 0.1          |
| U2 - dose-specific alpha   | 23.7   | 43.7   | 25.5   | 6.3    | 0.7    | 0.1          |
| U3 - shared alpha          | 23.7   | 45     | 24.1   | 6.4    | 0.7    | 0.1          |
| U3 - dose-specific alpha   | 24.4   | 44     | 23.7   | 7.1    | 0.7    | 0.1          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 21     | 47.2   | 12     | 1.5    | 0.2    | 18.1         |
| U2 - dose-specific alpha   | 21.1   | 46.4   | 12.1   | 2.1    | 0.2    | 18.1         |
| U3 - shared alpha          | 25.4   | 46     | 9.4    | 1.2    | 0      | 18           |
| U3 - dose-specific alpha   | 26.7   | 45     | 9      | 1.7    | 0      | 17.6         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 7.7    | 10.6   | 5.8    | 2      | 0.7    | -            |
| U2 - dose-specific alpha   | 7.7    | 10.6   | 5.9    | 1.9    | 0.7    | -            |
| U3 - shared alpha          | 7.7    | 10.6   | 5.9    | 1.9    | 0.7    | -            |
| U3 - dose-specific alpha   | 7.7    | 10.7   | 5.8    | 1.9    | 0.7    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 1.9    | 4.9    | 1.8    | 0.8    | 0.2    | -            |
| U2 - dose-specific alpha   | 1.9    | 4.9    | 1.8    | 0.8    | 0.2    | -            |
| U3 - shared alpha          | 1.9    | 4.9    | 1.8    | 0.8    | 0.2    | -            |
| U3 - dose-specific alpha   | 1.9    | 5      | 1.8    | 0.8    | 0.2    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 940.9        |
| U2 - dose-specific alpha   |        |        |        |        |        | 940.5        |
| U3 - shared alpha          |        |        |        |        |        | 939.8        |
| U3 - dose-specific alpha   |        |        |        |        |        | 940.7        |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 1                 |        |        |        |        |        |              |
| DLT rate                   | 0.30   | 0.35   | 0.40   | 0.45   | 0.50   |              |
| Efficacy rate              | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |              |
| Utility 2                  | -0.2   | -0.3   | -0.3   | -0.4   | -0.4   |              |
| Utility 3                  | 31     | 29.6   | 28.2   | 26.8   | 25.4   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 63.3   | 18.3   | 5.8    | 1.4    | 0.1    | 11.1         |
| U2 - dose-specific alpha   | 63.2   | 18.7   | 5.4    | 1.4    | 0.1    | 11.2         |
| U3 - shared alpha          | 63.2   | 18.4   | 5.5    | 1.4    | 0.1    | 11.4         |
| U3 - dose-specific alpha   | 62.5   | 19.5   | 5.3    | 1.3    | 0.1    | 11.3         |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 15.2   | 8      | 1.6    | 0.4    | 0.1    | 74.7         |
| U2 - dose-specific alpha   | 16     | 7.6    | 1.6    | 0.2    | 0.1    | 74.5         |
| U3 - shared alpha          | 15     | 9.3    | 1.2    | 0.1    | 0      | 74.4         |
| U3 - dose-specific alpha   | 14.9   | 9.3    | 1.3    | 0.1    | 0      | 74.4         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 11.2   | 5.3    | 1.9    | 1      | 0.2    | -            |
| U2 - dose-specific alpha   | 11.1   | 5.3    | 1.9    | 1      | 0.2    | -            |
| U3 - shared alpha          | 11.3   | 5.2    | 1.9    | 1      | 0.2    | -            |
| U3 - dose-specific alpha   | 11.3   | 5.2    | 1.9    | 0.9    | 0.2    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 3      | 2.8    | 0.6    | 0.4    | 0.1    | -            |
| U2 - dose-specific alpha   | 3      | 2.8    | 0.6    | 0.4    | 0.1    | -            |
| U3 - shared alpha          | 3.1    | 2.7    | 0.6    | 0.4    | 0.1    | -            |
| U3 - dose-specific alpha   | 3.1    | 2.7    | 0.6    | 0.4    | 0.1    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 688.8        |
| U2 - dose-specific alpha   |        |        |        |        |        | 688.5        |
| U3 - shared alpha          |        |        |        |        |        | 686.5        |
| U3 - dose-specific alpha   |        |        |        |        |        | 687.8        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 2                 |      |      |      |      |      |        |
| DLT rate                   | 0.15 | 0.20 | 0.30 | 0.35 | 0.45 |        |
| Efficacy rate              | 0.20 | 0.25 | 0.50 | 0.50 | 0.50 |        |
| Utility 2                  | 0.1  | 0.1  | 0.2  | 0.1  | 0.1  |        |
| Utility 3                  | 46   | 47   | 58   | 56   | 52   |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 7.8  | 31.3 | 41.6 | 16.6 | 2.4  | 0.3    |
| U2 - dose-specific alpha   | 7.5  | 31.7 | 41   | 17.1 | 2.4  | 0.3    |
| U3 - shared alpha          | 7.6  | 32.8 | 41.8 | 15.6 | 1.9  | 0.3    |
| U3 - dose-specific alpha   | 7.8  | 33   | 41.5 | 15.5 | 1.9  | 0.3    |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 26.9 | 19.4 | 38.3 | 11.5 | 1    | 2.9    |
| U2 - dose-specific alpha   | 27.1 | 20.1 | 37.4 | 11.4 | 1.1  | 2.9    |
| U3 - shared alpha          | 31.1 | 19.8 | 35.2 | 10.2 | 0.9  | 2.8    |
| U3 - dose-specific alpha   | 30.9 | 20.6 | 34.7 | 10   | 0.9  | 2.9    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 9.6  | 7.6  | 8.8  | 3.1  | 0.5  | -      |
| U2 - dose-specific alpha   | 9.6  | 7.5  | 8.8  | 3.1  | 0.5  | -      |
| U3 - shared alpha          | 10.1 | 7.7  | 8.5  | 2.8  | 0.4  | -      |
| U3 - dose-specific alpha   | 10.3 | 7.7  | 8.4  | 2.8  | 0.4  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 2.4  | 3.4  | 2.4  | 0.9  | 0.1  | -      |
| U2 - dose-specific alpha   | 2.4  | 3.4  | 2.4  | 0.9  | 0.1  | -      |
| U3 - shared alpha          | 2.6  | 3.5  | 2.4  | 0.9  | 0.1  | -      |
| U3 - dose-specific alpha   | 2.7  | 3.5  | 2.3  | 0.9  | 0.1  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1113   |
| U2 - dose-specific alpha   |      |      |      |      |      | 1115.2 |
| U3 - shared alpha          |      |      |      |      |      | 1111.1 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1110.9 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 3                 |        |        |        |        |        |              |
| DLT rate                   | 0.15   | 0.20   | 0.30   | 0.35   | 0.45   |              |
| Efficacy rate              | 0.10   | 0.40   | 0.40   | 0.40   | 0.40   |              |
| Utility 2                  | -0.1   | 0.2    | 0.1    | 0.1    | -0.1   |              |
| Utility 3                  | 40     | 56     | 52     | 50     | 46     |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 11.3   | 31.1   | 36.1   | 17.8   | 3.4    | 0.3          |
| U2 - dose-specific alpha   | 10.9   | 31.8   | 36.4   | 16.8   | 3.8    | 0.3          |
| U3 - shared alpha          | 11.1   | 35.6   | 34.6   | 15.3   | 3.1    | 0.3          |
| U3 - dose-specific alpha   | 11.3   | 35     | 35.4   | 15.2   | 2.8    | 0.3          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 8      | 47     | 24.8   | 9.7    | 1.3    | 9.2          |
| U2 - dose-specific alpha   | 8.2    | 47.6   | 24.3   | 9.6    | 1.4    | 8.9          |
| U3 - shared alpha          | 9.2    | 51.6   | 21.9   | 7.2    | 1      | 9.1          |
| U3 - dose-specific alpha   | 9.3    | 53.2   | 20.5   | 6.8    | 1      | 9.2          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 6.8    | 11.6   | 6.6    | 3      | 0.7    | -            |
| U2 - dose-specific alpha   | 6.8    | 11.7   | 6.5    | 3      | 0.7    | -            |
| U3 - shared alpha          | 6.9    | 12.4   | 6.1    | 2.7    | 0.5    | -            |
| U3 - dose-specific alpha   | 6.9    | 12.5   | 6.1    | 2.7    | 0.5    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 1.3    | 4.7    | 2      | 1      | 0.2    | -            |
| U2 - dose-specific alpha   | 1.3    | 4.8    | 1.9    | 1      | 0.2    | -            |
| U3 - shared alpha          | 1.4    | 5      | 1.8    | 0.9    | 0.2    | -            |
| U3 - dose-specific alpha   | 1.3    | 5.1    | 1.8    | 0.9    | 0.2    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1070         |
| U2 - dose-specific alpha   |        |        |        |        |        | 1069.8       |
| U3 - shared alpha          |        |        |        |        |        | 1065.9       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1066.2       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 4                 |      |      |      |      |      |        |
| DLT rate                   | 0.05 | 0.10 | 0.18 | 0.30 | 0.40 |        |
| Efficacy rate              | 0.05 | 0.10 | 0.25 | 0.50 | 0.55 |        |
| Utility 2                  | 0    | 0    | 0.1  | 0.2  | 0.1  |        |
| Utility 3                  | 41   | 42   | 47.8 | 58   | 57   |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0.8  | 2.8  | 18.9 | 58.6 | 18.9 | 0      |
| U2 - dose-specific alpha   | 0.8  | 2.5  | 18.7 | 59.7 | 18.3 | 0      |
| U3 - shared alpha          | 0.8  | 2.8  | 19.4 | 61   | 16   | 0      |
| U3 - dose-specific alpha   | 0.8  | 2.7  | 19.3 | 60.6 | 16.6 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 6.7  | 5.4  | 18.8 | 55.1 | 12.4 | 1.6    |
| U2 - dose-specific alpha   | 6.3  | 5.7  | 19.3 | 54.2 | 12.8 | 1.7    |
| U3 - shared alpha          | 8.6  | 6.8  | 19.4 | 54   | 9.6  | 1.6    |
| U3 - dose-specific alpha   | 8.4  | 7.5  | 18.9 | 53.5 | 10.1 | 1.6    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 4.2  | 4.7  | 7.1  | 10.7 | 3.1  | -      |
| U2 - dose-specific alpha   | 4.2  | 4.7  | 7.1  | 10.8 | 3    | -      |
| U3 - shared alpha          | 4.4  | 4.8  | 7.2  | 10.8 | 2.7  | -      |
| U3 - dose-specific alpha   | 4.3  | 4.8  | 7.2  | 10.8 | 2.7  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.5  | 3.1  | 2.1  | 3.4  | 0.9  | -      |
| U2 - dose-specific alpha   | 0.5  | 3.1  | 2.1  | 3.4  | 0.8  | -      |
| U3 - shared alpha          | 0.6  | 3.1  | 2.1  | 3.4  | 0.7  | -      |
| U3 - dose-specific alpha   | 0.6  | 3.1  | 2.1  | 3.4  | 0.7  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1098.6 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1098.5 |
| U3 - shared alpha          |      |      |      |      |      | 1093.2 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1093.9 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 5                 |        |        |        |        |        |              |
| DLT rate                   | 0.07   | 0.12   | 0.17   | 0.22   | 0.30   |              |
| Efficacy rate              | 0.05   | 0.30   | 0.30   | 0.30   | 0.30   |              |
| Utility 2                  | -0     | 0.2    | 0.1    | 0.1    | 0      |              |
| Utility 3                  | 40.2   | 53.2   | 51.2   | 49.2   | 46     |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 1.5    | 3.7    | 16.9   | 33.5   | 44.4   | 0            |
| U2 - dose-specific alpha   | 1.5    | 3.6    | 17.7   | 33     | 44.2   | 0            |
| U3 - shared alpha          | 1.5    | 6      | 19.2   | 34.6   | 38.7   | 0            |
| U3 - dose-specific alpha   | 1.5    | 5.4    | 18.8   | 33.7   | 40.6   | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 5.2    | 33.4   | 26.4   | 23.1   | 10.4   | 1.5          |
| U2 - dose-specific alpha   | 5.4    | 33.8   | 26.1   | 22.9   | 10.3   | 1.5          |
| U3 - shared alpha          | 6.9    | 44.3   | 23.2   | 18.4   | 5.7    | 1.5          |
| U3 - dose-specific alpha   | 7.5    | 42.1   | 23.3   | 19     | 6.6    | 1.5          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 4      | 7.5    | 7.2    | 7.2    | 3.9    | -            |
| U2 - dose-specific alpha   | 4      | 7.5    | 7.2    | 7.1    | 3.9    | -            |
| U3 - shared alpha          | 4.1    | 8.2    | 7.4    | 6.8    | 3.3    | -            |
| U3 - dose-specific alpha   | 4.1    | 8.2    | 7.3    | 6.9    | 3.3    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 0.3    | 3.9    | 2.3    | 2.7    | 1.4    | -            |
| U2 - dose-specific alpha   | 0.3    | 3.9    | 2.3    | 2.7    | 1.4    | -            |
| U3 - shared alpha          | 0.4    | 4.2    | 2.4    | 2.6    | 1.2    | -            |
| U3 - dose-specific alpha   | 0.4    | 4.2    | 2.4    | 2.6    | 1.2    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1047.9       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1047.7       |
| U3 - shared alpha          |        |        |        |        |        | 1046.7       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1046.3       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 6                 |      |      |      |      |      |        |
| DLT rate                   | 0.29 | 0.34 | 0.38 | 0.43 | 0.52 |        |
| Efficacy rate              | 0.30 | 0.32 | 0.35 | 0.38 | 0.40 |        |
| Utility 2                  | 0    | -0   | -0   | -0.1 | -0.1 |        |
| Utility 3                  | 46.4 | 45.6 | 45.8 | 45.6 | 43.2 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 53.1 | 23   | 8.3  | 2.4  | 0.2  | 13     |
| U2 - dose-specific alpha   | 52.8 | 23.5 | 8.7  | 2.1  | 0.2  | 12.7   |
| U3 - shared alpha          | 53.8 | 23.5 | 7.3  | 2.1  | 0.2  | 13.1   |
| U3 - dose-specific alpha   | 53.1 | 23.9 | 7.4  | 2.1  | 0.2  | 13.3   |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 66   | 12.8 | 3.6  | 1.1  | 0.2  | 16.3   |
| U2 - dose-specific alpha   | 65.2 | 13.5 | 3.6  | 1.1  | 0.2  | 16.4   |
| U3 - shared alpha          | 66.4 | 12.4 | 3    | 1.1  | 0.2  | 16.9   |
| U3 - dose-specific alpha   | 67.4 | 12.5 | 3    | 0.9  | 0.2  | 16     |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 18.3 | 5.7  | 2.2  | 0.8  | 0.2  | -      |
| U2 - dose-specific alpha   | 18.3 | 5.7  | 2.2  | 0.8  | 0.2  | -      |
| U3 - shared alpha          | 18.7 | 5.6  | 1.9  | 0.7  | 0.2  | -      |
| U3 - dose-specific alpha   | 18.9 | 5.6  | 2    | 0.7  | 0.2  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 4.7  | 2.3  | 0.7  | 0.3  | 0.1  | -      |
| U2 - dose-specific alpha   | 4.7  | 2.3  | 0.7  | 0.3  | 0.1  | -      |
| U3 - shared alpha          | 4.9  | 2.3  | 0.6  | 0.3  | 0.1  | -      |
| U3 - dose-specific alpha   | 4.9  | 2.3  | 0.6  | 0.3  | 0.1  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1059   |
| U2 - dose-specific alpha   |      |      |      |      |      | 1058.4 |
| U3 - shared alpha          |      |      |      |      |      | 1054.9 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1059.8 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 7                 |        |        |        |        |        |              |
| DLT rate                   | 0.16   | 0.20   | 0.26   | 0.31   | 0.36   |              |
| Efficacy rate              | 0.20   | 0.50   | 0.50   | 0.50   | 0.50   |              |
| Utility 2                  | 0      | 0.3    | 0.2    | 0.2    | 0.1    |              |
| Utility 3                  | 45.6   | 62     | 59.6   | 57.6   | 55.6   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 8.8    | 46     | 29.9   | 11.7   | 3.1    | 0.5          |
| U2 - dose-specific alpha   | 8.6    | 46.5   | 29.1   | 12.2   | 3.2    | 0.4          |
| U3 - shared alpha          | 8.7    | 53.1   | 24.7   | 10.5   | 2.6    | 0.4          |
| U3 - dose-specific alpha   | 8.7    | 52.6   | 25.5   | 10.3   | 2.5    | 0.4          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 12.4   | 51     | 24.4   | 7.8    | 1.5    | 2.9          |
| U2 - dose-specific alpha   | 12.2   | 51.6   | 23.5   | 8      | 1.8    | 2.9          |
| U3 - shared alpha          | 14.2   | 55.8   | 19     | 6.8    | 1.2    | 3            |
| U3 - dose-specific alpha   | 14     | 55.1   | 20.1   | 6.8    | 1.2    | 2.8          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 7.9    | 13.7   | 5.6    | 2      | 0.4    | -            |
| U2 - dose-specific alpha   | 7.9    | 13.5   | 5.7    | 2      | 0.4    | -            |
| U3 - shared alpha          | 8.2    | 14.2   | 5      | 1.8    | 0.3    | -            |
| U3 - dose-specific alpha   | 8.3    | 14.2   | 5      | 1.8    | 0.3    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 1.6    | 4.9    | 1.6    | 0.6    | 0.1    | -            |
| U2 - dose-specific alpha   | 1.6    | 4.9    | 1.6    | 0.6    | 0.1    | -            |
| U3 - shared alpha          | 1.7    | 5.1    | 1.3    | 0.6    | 0.1    | -            |
| U3 - dose-specific alpha   | 1.8    | 5.1    | 1.3    | 0.6    | 0.1    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1145.7       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1145.8       |
| U3 - shared alpha          |        |        |        |        |        | 1143.1       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1145         |

|                            |      |      |      |      |      |       |
|----------------------------|------|------|------|------|------|-------|
| Scenario 8                 |      |      |      |      |      |       |
| DLT rate                   | 0.32 | 0.37 | 0.42 | 0.48 | 0.52 |       |
| Efficacy rate              | 0.10 | 0.15 | 0.20 | 0.30 | 0.40 |       |
| Utility 2                  | -0.2 | -0.2 | -0.2 | -0.2 | -0.1 |       |
| Utility 3                  | 33.2 | 34.2 | 35.2 | 38.8 | 43.2 |       |
| MTD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 62.9 | 13.9 | 4    | 1.1  | 0.3  | 17.8  |
| U2 - dose-specific alpha   | 63.7 | 13.5 | 4.4  | 0.9  | 0.3  | 17.2  |
| U3 - shared alpha          | 63   | 13.8 | 4.1  | 0.9  | 0.3  | 17.9  |
| U3 - dose-specific alpha   | 63.7 | 14.1 | 3.8  | 0.8  | 0.3  | 17.3  |
| OBD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 29.4 | 8.3  | 1.7  | 0.4  | 0.2  | 60    |
| U2 - dose-specific alpha   | 28.3 | 8.9  | 1.6  | 0.3  | 0.2  | 60.7  |
| U3 - shared alpha          | 29.4 | 8.1  | 1.6  | 0.3  | 0.2  | 60.4  |
| U3 - dose-specific alpha   | 29.2 | 8.5  | 1.3  | 0.3  | 0.2  | 60.5  |
| Mean administrations       |      |      |      |      |      |       |
| U2 - shared alpha          | 13.4 | 5.3  | 1.8  | 0.8  | 0.1  | -     |
| U2 - dose-specific alpha   | 13.3 | 5.4  | 1.8  | 0.8  | 0.2  | -     |
| U3 - shared alpha          | 13.6 | 5.2  | 1.6  | 0.7  | 0.1  | -     |
| U3 - dose-specific alpha   | 13.6 | 5.2  | 1.6  | 0.7  | 0.1  | -     |
| Mean IPDE administrations  |      |      |      |      |      |       |
| U2 - shared alpha          | 3.6  | 2.6  | 0.6  | 0.3  | 0.1  | -     |
| U2 - dose-specific alpha   | 3.6  | 2.6  | 0.6  | 0.3  | 0.1  | -     |
| U3 - shared alpha          | 3.7  | 2.5  | 0.5  | 0.3  | 0    | -     |
| U3 - dose-specific alpha   | 3.7  | 2.6  | 0.5  | 0.3  | 0    | -     |
| Mean trial duration (days) |      |      |      |      |      | Days  |
| U2 - shared alpha          |      |      |      |      |      | 775.8 |
| U2 - dose-specific alpha   |      |      |      |      |      | 774.6 |
| U3 - shared alpha          |      |      |      |      |      | 773.1 |
| U3 - dose-specific alpha   |      |      |      |      |      | 774.9 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 9                 |        |        |        |        |        |              |
| DLT rate                   | 0.15   | 0.18   | 0.22   | 0.26   | 0.29   |              |
| Efficacy rate              | 0.05   | 0.08   | 0.20   | 0.25   | 0.50   |              |
| Utility 2                  | -0.1   | -0.1   | -0     | -0     | 0.2    |              |
| Utility 3                  | 37     | 37.6   | 43.2   | 44.6   | 58.4   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 11.7   | 12.6   | 19.3   | 26.7   | 29.4   | 0.3          |
| U2 - dose-specific alpha   | 11.4   | 13.5   | 18.5   | 26.9   | 29.4   | 0.3          |
| U3 - shared alpha          | 12.1   | 13.4   | 18.3   | 27.5   | 28.4   | 0.3          |
| U3 - dose-specific alpha   | 11.5   | 14.3   | 18     | 28     | 27.9   | 0.3          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 11.6   | 11.7   | 21.5   | 17.5   | 21.8   | 15.9         |
| U2 - dose-specific alpha   | 12.3   | 12.1   | 20.8   | 16.4   | 22.1   | 16.3         |
| U3 - shared alpha          | 15.5   | 13.9   | 19.7   | 16     | 17.8   | 17.1         |
| U3 - dose-specific alpha   | 15.9   | 14.5   | 19.9   | 15.2   | 17.8   | 16.7         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 6.7    | 5.8    | 6      | 5.2    | 4.2    | -            |
| U2 - dose-specific alpha   | 6.7    | 5.9    | 6      | 5.1    | 4.2    | -            |
| U3 - shared alpha          | 6.9    | 5.9    | 6      | 5.2    | 3.9    | -            |
| U3 - dose-specific alpha   | 6.9    | 6      | 6      | 5.1    | 3.9    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 1.5    | 3.4    | 2      | 2.1    | 1.3    | -            |
| U2 - dose-specific alpha   | 1.5    | 3.4    | 2      | 2      | 1.3    | -            |
| U3 - shared alpha          | 1.6    | 3.4    | 2      | 2      | 1.2    | -            |
| U3 - dose-specific alpha   | 1.6    | 3.4    | 2      | 2      | 1.2    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 977.8        |
| U2 - dose-specific alpha   |        |        |        |        |        | 978          |
| U3 - shared alpha          |        |        |        |        |        | 974.9        |
| U3 - dose-specific alpha   |        |        |        |        |        | 974.7        |

|                            |      |      |      |      |      |       |
|----------------------------|------|------|------|------|------|-------|
| Scenario 10                |      |      |      |      |      |       |
| DLT rate                   | 0.23 | 0.28 | 0.32 | 0.39 | 0.43 |       |
| Efficacy rate              | 0.05 | 0.40 | 0.42 | 0.45 | 0.50 |       |
| Utility 2                  | -0.2 | 0.1  | 0.1  | 0.1  | 0.1  |       |
| Utility 3                  | 33.8 | 52.8 | 52.4 | 51.4 | 52.8 |       |
| MTD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 44.2 | 25.9 | 18.6 | 7.3  | 1.3  | 2.7   |
| U2 - dose-specific alpha   | 44.2 | 25.6 | 19.3 | 6.9  | 1.2  | 2.8   |
| U3 - shared alpha          | 44   | 27.7 | 18   | 6.6  | 1.1  | 2.6   |
| U3 - dose-specific alpha   | 43.8 | 28.2 | 17.8 | 6.5  | 1.1  | 2.6   |
| OBD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 7.6  | 31.5 | 14.2 | 4    | 0.9  | 41.8  |
| U2 - dose-specific alpha   | 7.6  | 32.5 | 13.5 | 3.6  | 0.9  | 41.9  |
| U3 - shared alpha          | 8.1  | 33.7 | 12.5 | 3.4  | 0.8  | 41.5  |
| U3 - dose-specific alpha   | 7.9  | 34   | 12.7 | 3.2  | 0.8  | 41.4  |
| Mean administrations       |      |      |      |      |      |       |
| U2 - shared alpha          | 8.6  | 9.6  | 4    | 1.6  | 0.4  | -     |
| U2 - dose-specific alpha   | 8.6  | 9.6  | 4    | 1.6  | 0.4  | -     |
| U3 - shared alpha          | 8.7  | 9.9  | 3.8  | 1.5  | 0.3  | -     |
| U3 - dose-specific alpha   | 8.7  | 9.9  | 3.9  | 1.4  | 0.3  | -     |
| Mean IPDE administrations  |      |      |      |      |      |       |
| U2 - shared alpha          | 1.8  | 4.1  | 1.1  | 0.5  | 0.1  | -     |
| U2 - dose-specific alpha   | 1.8  | 4.1  | 1.1  | 0.6  | 0.1  | -     |
| U3 - shared alpha          | 1.9  | 4.2  | 1.1  | 0.5  | 0.1  | -     |
| U3 - dose-specific alpha   | 1.9  | 4.2  | 1.1  | 0.5  | 0.1  | -     |
| Mean trial duration (days) |      |      |      |      |      | Days  |
| U2 - shared alpha          |      |      |      |      |      | 903.3 |
| U2 - dose-specific alpha   |      |      |      |      |      | 902   |
| U3 - shared alpha          |      |      |      |      |      | 900.6 |
| U3 - dose-specific alpha   |      |      |      |      |      | 901.6 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 11                |        |        |        |        |        |              |
| DLT rate                   | 0.22   | 0.32   | 0.44   | 0.59   | 0.67   |              |
| Efficacy rate              | 0.20   | 0.20   | 0.20   | 0.20   | 0.20   |              |
| Utility 2                  | -0     | -0.1   | -0.2   | -0.4   | -0.5   |              |
| Utility 3                  | 43.2   | 39.2   | 34.4   | 28.4   | 25.2   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 41.3   | 43.1   | 11.6   | 0.7    | 0      | 3.3          |
| U2 - dose-specific alpha   | 40.5   | 43.3   | 11.9   | 1      | 0      | 3.3          |
| U3 - shared alpha          | 41.7   | 43.4   | 10.7   | 0.8    | 0      | 3.4          |
| U3 - dose-specific alpha   | 40.9   | 43.9   | 11.4   | 0.5    | 0      | 3.3          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 60.9   | 21.5   | 3.3    | 0.1    | 0      | 14.2         |
| U2 - dose-specific alpha   | 60.4   | 20.9   | 4.2    | 0.2    | 0      | 14.3         |
| U3 - shared alpha          | 62.6   | 20.6   | 2.4    | 0.1    | 0      | 14.3         |
| U3 - dose-specific alpha   | 62.6   | 20.4   | 2.7    | 0.1    | 0      | 14.2         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 16.8   | 7.5    | 2.7    | 1      | 0.1    | -            |
| U2 - dose-specific alpha   | 16.8   | 7.4    | 2.7    | 1      | 0.1    | -            |
| U3 - shared alpha          | 17.2   | 7.4    | 2.5    | 0.9    | 0.1    | -            |
| U3 - dose-specific alpha   | 17.2   | 7.4    | 2.6    | 0.9    | 0.1    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 4.9    | 3.4    | 0.8    | 0.4    | 0      | -            |
| U2 - dose-specific alpha   | 4.9    | 3.4    | 0.9    | 0.4    | 0      | -            |
| U3 - shared alpha          | 5      | 3.4    | 0.8    | 0.3    | 0      | -            |
| U3 - dose-specific alpha   | 5      | 3.4    | 0.8    | 0.3    | 0      | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1022.3       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1021.9       |
| U3 - shared alpha          |        |        |        |        |        | 1020.2       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1021.8       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 12                |      |      |      |      |      |        |
| DLT rate                   | 0.06 | 0.09 | 0.18 | 0.29 | 0.44 |        |
| Efficacy rate              | 0.05 | 0.10 | 0.12 | 0.40 | 0.60 |        |
| Utility 2                  | -0   | 0    | -0.1 | 0.1  | 0.2  |        |
| Utility 3                  | 40.6 | 42.4 | 40   | 52.4 | 58.4 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 1.1  | 2.6  | 17.8 | 56.6 | 21.9 | 0      |
| U2 - dose-specific alpha   | 1.2  | 2.7  | 17.7 | 55.9 | 22.5 | 0      |
| U3 - shared alpha          | 1.1  | 2.7  | 18   | 56.8 | 21.4 | 0      |
| U3 - dose-specific alpha   | 1.1  | 2.7  | 18.2 | 57.7 | 20.3 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 11.4 | 10.8 | 11.9 | 47   | 16.4 | 2.5    |
| U2 - dose-specific alpha   | 11.4 | 12.4 | 10.9 | 46.1 | 16.7 | 2.5    |
| U3 - shared alpha          | 15.3 | 13.8 | 9.2  | 43.7 | 15.5 | 2.5    |
| U3 - dose-specific alpha   | 15.6 | 14.8 | 9.5  | 43.7 | 13.9 | 2.5    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 4.6  | 5.3  | 5.9  | 9.5  | 4.5  | -      |
| U2 - dose-specific alpha   | 4.6  | 5.3  | 5.9  | 9.5  | 4.5  | -      |
| U3 - shared alpha          | 4.8  | 5.4  | 5.8  | 9.5  | 4.2  | -      |
| U3 - dose-specific alpha   | 4.9  | 5.5  | 5.8  | 9.4  | 4.2  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.7  | 3.3  | 1.7  | 3.4  | 1.2  | -      |
| U2 - dose-specific alpha   | 0.7  | 3.3  | 1.7  | 3.3  | 1.2  | -      |
| U3 - shared alpha          | 0.8  | 3.4  | 1.7  | 3.4  | 1.1  | -      |
| U3 - dose-specific alpha   | 0.8  | 3.4  | 1.7  | 3.3  | 1.1  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1071.1 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1071.4 |
| U3 - shared alpha          |      |      |      |      |      | 1068.5 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1067.3 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 13                |        |        |        |        |        |              |
| DLT rate                   | 0.21   | 0.29   | 0.30   | 0.39   | 0.46   |              |
| Efficacy rate              | 0.05   | 0.30   | 0.32   | 0.35   | 0.40   |              |
| Utility 2                  | -0.2   | 0      | 0      | -0     | -0.1   |              |
| Utility 3                  | 34.6   | 46.4   | 47.2   | 45.4   | 45.6   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 40.9   | 25.1   | 18.7   | 11.1   | 2.2    | 2            |
| U2 - dose-specific alpha   | 40.8   | 25.1   | 19.1   | 11     | 2      | 2            |
| U3 - shared alpha          | 41     | 25.9   | 19.2   | 10.4   | 1.5    | 2            |
| U3 - dose-specific alpha   | 41.4   | 25.6   | 19.1   | 10.2   | 1.8    | 1.9          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 9      | 32     | 14.2   | 4.8    | 1.2    | 38.8         |
| U2 - dose-specific alpha   | 9.4    | 32.5   | 14.2   | 4.1    | 1.2    | 38.6         |
| U3 - shared alpha          | 10.5   | 32.6   | 13.8   | 3.5    | 0.8    | 38.8         |
| U3 - dose-specific alpha   | 10.5   | 32.9   | 12.4   | 4.1    | 0.9    | 39.2         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 8.6    | 8.8    | 4.5    | 2.3    | 0.6    | -            |
| U2 - dose-specific alpha   | 8.5    | 8.8    | 4.5    | 2.2    | 0.5    | -            |
| U3 - shared alpha          | 8.7    | 9.1    | 4.3    | 2.1    | 0.5    | -            |
| U3 - dose-specific alpha   | 8.7    | 9.1    | 4.3    | 2.1    | 0.5    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 1.9    | 4.1    | 1.4    | 0.8    | 0.2    | -            |
| U2 - dose-specific alpha   | 1.9    | 4.1    | 1.4    | 0.8    | 0.2    | -            |
| U3 - shared alpha          | 2      | 4.2    | 1.3    | 0.8    | 0.2    | -            |
| U3 - dose-specific alpha   | 2      | 4.2    | 1.3    | 0.8    | 0.2    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 894.6        |
| U2 - dose-specific alpha   |        |        |        |        |        | 893          |
| U3 - shared alpha          |        |        |        |        |        | 890.5        |
| U3 - dose-specific alpha   |        |        |        |        |        | 891.4        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 14                |      |      |      |      |      |        |
| DLT rate                   | 0.09 | 0.15 | 0.23 | 0.28 | 0.39 |        |
| Efficacy rate              | 0.05 | 0.08 | 0.12 | 0.30 | 0.40 |        |
| Utility 2                  | -0   | -0.1 | -0.1 | 0    | 0    |        |
| Utility 3                  | 39.4 | 38.8 | 38   | 46.8 | 48.4 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 2.8  | 10.3 | 25   | 38.9 | 23   | 0      |
| U2 - dose-specific alpha   | 2.8  | 9.8  | 25.7 | 38.5 | 23.2 | 0      |
| U3 - shared alpha          | 2.9  | 10.1 | 25.5 | 39.9 | 21.6 | 0      |
| U3 - dose-specific alpha   | 2.7  | 9.5  | 25.5 | 39.8 | 22.5 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 16.3 | 17.2 | 15.7 | 31.7 | 12.3 | 6.8    |
| U2 - dose-specific alpha   | 15.6 | 16.7 | 16.9 | 31.4 | 12.6 | 6.8    |
| U3 - shared alpha          | 23   | 20.1 | 14.6 | 27   | 8.3  | 7      |
| U3 - dose-specific alpha   | 21.5 | 19.9 | 15.5 | 27.7 | 8.6  | 6.8    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 6    | 6.4  | 5.9  | 7.4  | 3.6  | -      |
| U2 - dose-specific alpha   | 6    | 6.4  | 6    | 7.3  | 3.7  | -      |
| U3 - shared alpha          | 6.4  | 6.5  | 5.8  | 7.3  | 3.4  | -      |
| U3 - dose-specific alpha   | 6.4  | 6.4  | 5.8  | 7.3  | 3.4  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 1.3  | 3.7  | 1.9  | 2.8  | 1.2  | -      |
| U2 - dose-specific alpha   | 1.3  | 3.7  | 1.9  | 2.8  | 1.2  | -      |
| U3 - shared alpha          | 1.4  | 3.8  | 1.8  | 2.8  | 1.1  | -      |
| U3 - dose-specific alpha   | 1.5  | 3.8  | 1.9  | 2.8  | 1.1  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1014.7 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1014.6 |
| U3 - shared alpha          |      |      |      |      |      | 1014.2 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1014.1 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 15                |        |        |        |        |        |              |
| DLT rate                   | 0.13   | 0.20   | 0.32   | 0.39   | 0.46   |              |
| Efficacy rate              | 0.10   | 0.10   | 0.10   | 0.10   | 0.10   |              |
| Utility 2                  | -0     | -0.1   | -0.2   | -0.3   | -0.4   |              |
| Utility 3                  | 40.8   | 38     | 33.2   | 30.4   | 27.6   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 9      | 29.6   | 36.3   | 19.8   | 5.1    | 0.2          |
| U2 - dose-specific alpha   | 8.9    | 29.2   | 37.1   | 20.3   | 4.3    | 0.2          |
| U3 - shared alpha          | 8.9    | 30.3   | 35.7   | 20.3   | 4.6    | 0.2          |
| U3 - dose-specific alpha   | 9      | 30     | 35.5   | 20.4   | 4.9    | 0.2          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 41.2   | 29     | 11.8   | 4.9    | 0.5    | 12.6         |
| U2 - dose-specific alpha   | 40     | 29.4   | 11.9   | 5.2    | 0.4    | 13.1         |
| U3 - shared alpha          | 45.7   | 30.6   | 7.3    | 3.5    | 0.1    | 12.8         |
| U3 - dose-specific alpha   | 46.1   | 28.6   | 8.2    | 3.7    | 0.3    | 13.1         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 9.7    | 8.5    | 5.8    | 3.6    | 1.2    | -            |
| U2 - dose-specific alpha   | 9.7    | 8.4    | 5.8    | 3.6    | 1.1    | -            |
| U3 - shared alpha          | 10.4   | 8.3    | 5.5    | 3.5    | 1.1    | -            |
| U3 - dose-specific alpha   | 10.4   | 8.3    | 5.4    | 3.5    | 1.1    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 2.8    | 4.3    | 2      | 1.5    | 0.4    | -            |
| U2 - dose-specific alpha   | 2.8    | 4.3    | 2      | 1.5    | 0.4    | -            |
| U3 - shared alpha          | 3.1    | 4.2    | 1.8    | 1.5    | 0.4    | -            |
| U3 - dose-specific alpha   | 3.1    | 4.3    | 1.8    | 1.5    | 0.4    | -            |
| Mean trial duration (days) |        |        |        |        |        |              |
| U2 - shared alpha          |        |        |        |        |        | Days         |
| U2 - dose-specific alpha   |        |        |        |        |        | 972.4        |
| U3 - shared alpha          |        |        |        |        |        | 971.4        |
| U3 - dose-specific alpha   |        |        |        |        |        | 971          |
|                            |        |        |        |        |        | 970.4        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 16                |      |      |      |      |      |        |
| DLT rate                   | 0.02 | 0.06 | 0.09 | 0.14 | 0.26 |        |
| Efficacy rate              | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 |        |
| Utility 2                  | 0.2  | 0.1  | 0.1  | 0.1  | -0.1 |        |
| Utility 3                  | 51.2 | 49.6 | 48.4 | 46.4 | 41.6 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0    | 1    | 1.9  | 11.7 | 85.4 | 0      |
| U2 - dose-specific alpha   | 0    | 1.6  | 2    | 10.9 | 85.5 | 0      |
| U3 - shared alpha          | 0    | 2.1  | 3.8  | 12.1 | 82   | 0      |
| U3 - dose-specific alpha   | 0    | 2.5  | 2.4  | 13   | 82.1 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 37.8 | 27.2 | 16   | 12.1 | 6.9  | 0      |
| U2 - dose-specific alpha   | 38.1 | 26.6 | 15.2 | 12.4 | 7.7  | 0      |
| U3 - shared alpha          | 42   | 30   | 13.7 | 10.2 | 4.1  | 0      |
| U3 - dose-specific alpha   | 43.9 | 27.9 | 14   | 10.4 | 3.8  | 0      |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 4.3  | 5    | 5.9  | 8.4  | 6.3  | -      |
| U2 - dose-specific alpha   | 4.4  | 5.1  | 5.9  | 8.3  | 6.2  | -      |
| U3 - shared alpha          | 4.7  | 5.4  | 6.2  | 8.1  | 5.7  | -      |
| U3 - dose-specific alpha   | 4.7  | 5.3  | 6.2  | 8.2  | 5.6  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.5  | 2.9  | 2.3  | 3.6  | 2.4  | -      |
| U2 - dose-specific alpha   | 0.6  | 2.9  | 2.3  | 3.6  | 2.3  | -      |
| U3 - shared alpha          | 0.7  | 3.1  | 2.3  | 3.5  | 2.1  | -      |
| U3 - dose-specific alpha   | 0.7  | 3    | 2.3  | 3.5  | 2.1  | -      |
| Mean trial duration (days) |      |      |      |      |      |        |
| U2 - shared alpha          |      |      |      |      |      | Days   |
| U2 - dose-specific alpha   |      |      |      |      |      | 1009.1 |
| U3 - shared alpha          |      |      |      |      |      | 1008.9 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1007.1 |
|                            |      |      |      |      |      | 1006.9 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 17                |        |        |        |        |        |              |
| DLT rate                   | 0.11   | 0.23   | 0.39   | 0.58   | 0.76   |              |
| Efficacy rate              | 0.30   | 0.50   | 0.50   | 0.50   | 0.50   |              |
| Utility 2                  | 0.2    | 0.3    | 0.1    | -0.1   | -0.3   |              |
| Utility 3                  | 53.6   | 60.8   | 54.4   | 46.8   | 39.6   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 5.3    | 66.4   | 25.9   | 2.3    | 0      | 0.1          |
| U2 - dose-specific alpha   | 6.2    | 65.3   | 26.5   | 1.9    | 0      | 0.1          |
| U3 - shared alpha          | 6      | 70.3   | 22     | 1.6    | 0      | 0.1          |
| U3 - dose-specific alpha   | 6      | 70.3   | 22     | 1.6    | 0      | 0.1          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 18.4   | 65.1   | 14.8   | 1.2    | 0      | 0.5          |
| U2 - dose-specific alpha   | 20.5   | 63.5   | 14.4   | 1.1    | 0      | 0.5          |
| U3 - shared alpha          | 20.9   | 65.7   | 12.1   | 0.6    | 0      | 0.7          |
| U3 - dose-specific alpha   | 22.3   | 65     | 11.4   | 0.7    | 0      | 0.6          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 9.4    | 15.4   | 4.4    | 0.7    | 0.1    | -            |
| U2 - dose-specific alpha   | 9.6    | 15.3   | 4.3    | 0.7    | 0.1    | -            |
| U3 - shared alpha          | 10     | 15.6   | 3.8    | 0.5    | 0      | -            |
| U3 - dose-specific alpha   | 10.2   | 15.5   | 3.6    | 0.5    | 0      | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 2.1    | 5.1    | 1.2    | 0.2    | 0      | -            |
| U2 - dose-specific alpha   | 2.1    | 5      | 1.2    | 0.2    | 0      | -            |
| U3 - shared alpha          | 2.3    | 5.2    | 1      | 0.2    | 0      | -            |
| U3 - dose-specific alpha   | 2.4    | 5.2    | 1      | 0.2    | 0      | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1180.1       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1178.3       |
| U3 - shared alpha          |        |        |        |        |        | 1173.4       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1171.7       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 18                |      |      |      |      |      |        |
| DLT rate                   | 0.02 | 0.04 | 0.10 | 0.20 | 0.35 |        |
| Efficacy rate              | 0.05 | 0.10 | 0.15 | 0.20 | 0.50 |        |
| Utility 2                  | 0    | 0.1  | 0.1  | 0    | 0.1  |        |
| Utility 3                  | 42.2 | 44.4 | 45   | 44   | 56   |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0.1  | 0.1  | 2.6  | 25.4 | 71.8 | 0      |
| U2 - dose-specific alpha   | 0.1  | 0.1  | 2.3  | 25.6 | 71.9 | 0      |
| U3 - shared alpha          | 0.1  | 0.1  | 2.4  | 25.2 | 72.2 | 0      |
| U3 - dose-specific alpha   | 0.1  | 0.1  | 2.7  | 26   | 71.1 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 10.3 | 10.2 | 8.8  | 15.7 | 54.9 | 0.1    |
| U2 - dose-specific alpha   | 9.9  | 10.8 | 8.9  | 16.4 | 53.9 | 0.1    |
| U3 - shared alpha          | 16.2 | 13.1 | 10.2 | 13.3 | 47.1 | 0.1    |
| U3 - dose-specific alpha   | 16.8 | 14.5 | 9.8  | 13.3 | 45.5 | 0.1    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 3.4  | 4.1  | 5.2  | 7.1  | 10.2 | -      |
| U2 - dose-specific alpha   | 3.5  | 4    | 5.2  | 7.1  | 10.2 | -      |
| U3 - shared alpha          | 3.5  | 4.2  | 5.4  | 7.2  | 9.7  | -      |
| U3 - dose-specific alpha   | 3.5  | 4.2  | 5.3  | 7.2  | 9.7  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.2  | 2.8  | 1.8  | 3    | 2.9  | -      |
| U2 - dose-specific alpha   | 0.2  | 2.8  | 1.8  | 2.9  | 2.9  | -      |
| U3 - shared alpha          | 0.2  | 2.9  | 1.9  | 3    | 2.8  | -      |
| U3 - dose-specific alpha   | 0.2  | 2.9  | 1.8  | 3    | 2.8  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1060.7 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1061.8 |
| U3 - shared alpha          |      |      |      |      |      | 1057.3 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1058.1 |

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 19                |        |        |        |        |        |              |
| DLT rate                   | 0.11   | 0.26   | 0.39   | 0.60   | 0.77   |              |
| Efficacy rate              | 0.08   | 0.30   | 0.32   | 0.35   | 0.40   |              |
| Utility 2                  | -0     | 0      | -0.1   | -0.2   | -0.4   |              |
| Utility 3                  | 40.4   | 47.6   | 43.6   | 37     | 33.2   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 18.1   | 47.3   | 30.4   | 4.2    | 0      | 0            |
| U2 - dose-specific alpha   | 18.4   | 46.4   | 31.4   | 3.8    | 0      | 0            |
| U3 - shared alpha          | 18     | 47.2   | 30.7   | 4.1    | 0      | 0            |
| U3 - dose-specific alpha   | 17.9   | 47.2   | 30.9   | 4      | 0      | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 11.7   | 56.2   | 16.7   | 1.2    | 0      | 14.2         |
| U2 - dose-specific alpha   | 11.8   | 57.2   | 15.8   | 0.7    | 0      | 14.5         |
| U3 - shared alpha          | 13.2   | 57.8   | 13.6   | 0.8    | 0      | 14.6         |
| U3 - dose-specific alpha   | 13.7   | 56.9   | 14.1   | 0.6    | 0      | 14.7         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 7.9    | 12.6   | 5.9    | 1.9    | 0.2    | -            |
| U2 - dose-specific alpha   | 7.9    | 12.6   | 5.9    | 1.9    | 0.2    | -            |
| U3 - shared alpha          | 8.2    | 12.8   | 5.6    | 1.7    | 0.2    | -            |
| U3 - dose-specific alpha   | 8.1    | 12.8   | 5.7    | 1.7    | 0.2    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 1.7    | 5.5    | 1.7    | 0.7    | 0.1    | -            |
| U2 - dose-specific alpha   | 1.7    | 5.5    | 1.7    | 0.7    | 0.1    | -            |
| U3 - shared alpha          | 1.8    | 5.6    | 1.6    | 0.6    | 0.1    | -            |
| U3 - dose-specific alpha   | 1.8    | 5.6    | 1.7    | 0.6    | 0.1    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1033.5       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1032.7       |
| U3 - shared alpha          |        |        |        |        |        | 1029.4       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1029.1       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 20                |      |      |      |      |      |        |
| DLT rate                   | 0.07 | 0.14 | 0.31 | 0.41 | 0.52 |        |
| Efficacy rate              | 0.05 | 0.25 | 0.25 | 0.25 | 0.25 |        |
| Utility 2                  | -0   | 0.1  | -0.1 | -0.2 | -0.3 |        |
| Utility 3                  | 40.2 | 49.4 | 42.6 | 38.6 | 34.2 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 2.2  | 14.8 | 49.3 | 28.6 | 5.1  | 0      |
| U2 - dose-specific alpha   | 2.3  | 14.4 | 49.6 | 28.8 | 4.9  | 0      |
| U3 - shared alpha          | 2.2  | 13.8 | 49.8 | 29.7 | 4.5  | 0      |
| U3 - dose-specific alpha   | 2.2  | 14.1 | 49.4 | 29.7 | 4.6  | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 8    | 49.8 | 29.1 | 9.7  | 0.9  | 2.5    |
| U2 - dose-specific alpha   | 8.4  | 48.8 | 30.1 | 8.9  | 1.3  | 2.5    |
| U3 - shared alpha          | 10.2 | 53   | 25.3 | 8.2  | 0.7  | 2.6    |
| U3 - dose-specific alpha   | 10.1 | 53.8 | 24.6 | 8.2  | 0.8  | 2.5    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 4.9  | 10.6 | 8.2  | 4.6  | 1.4  | -      |
| U2 - dose-specific alpha   | 4.8  | 10.5 | 8.3  | 4.6  | 1.4  | -      |
| U3 - shared alpha          | 5.1  | 11.1 | 7.8  | 4.4  | 1.3  | -      |
| U3 - dose-specific alpha   | 5.1  | 11.3 | 7.7  | 4.3  | 1.3  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.7  | 5    | 2.6  | 1.7  | 0.5  | -      |
| U2 - dose-specific alpha   | 0.7  | 5    | 2.6  | 1.7  | 0.5  | -      |
| U3 - shared alpha          | 0.8  | 5.2  | 2.5  | 1.6  | 0.4  | -      |
| U3 - dose-specific alpha   | 0.9  | 5.3  | 2.4  | 1.6  | 0.4  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1052.6 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1052.2 |
| U3 - shared alpha          |      |      |      |      |      | 1048.1 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1046.8 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 21                |        |        |        |        |        |              |
| DLT rate                   | 0.37   | 0.56   | 0.69   | 0.83   | 0.88   |              |
| Efficacy rate              | 0.50   | 0.50   | 0.50   | 0.50   | 0.50   |              |
| Utility 2                  | 0.1    | -0.1   | -0.2   | -0.3   | -0.4   |              |
| Utility 3                  | 55.2   | 47.6   | 42.4   | 36.8   | 34.8   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 54.3   | 2.3    | 0      | 0      | 0      | 43.4         |
| U2 - dose-specific alpha   | 54.4   | 2.3    | 0      | 0      | 0      | 43.3         |
| U3 - shared alpha          | 54.8   | 2.2    | 0      | 0      | 0      | 43           |
| U3 - dose-specific alpha   | 55.2   | 2.1    | 0      | 0      | 0      | 42.7         |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 52.5   | 1      | 0      | 0      | 0      | 46.5         |
| U2 - dose-specific alpha   | 52     | 1.1    | 0      | 0      | 0      | 46.9         |
| U3 - shared alpha          | 52.9   | 1      | 0      | 0      | 0      | 46.1         |
| U3 - dose-specific alpha   | 52.6   | 1      | 0      | 0      | 0      | 46.4         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 19.8   | 2.3    | 0.1    | 0      | 0      | -            |
| U2 - dose-specific alpha   | 19.8   | 2.3    | 0.1    | 0      | 0      | -            |
| U3 - shared alpha          | 20     | 2.2    | 0.1    | 0      | 0      | -            |
| U3 - dose-specific alpha   | 19.9   | 2.3    | 0.1    | 0      | 0      | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 3.8    | 0.9    | 0      | 0      | 0      | -            |
| U2 - dose-specific alpha   | 3.8    | 0.9    | 0      | 0      | 0      | -            |
| U3 - shared alpha          | 3.8    | 0.9    | 0      | 0      | 0      | -            |
| U3 - dose-specific alpha   | 3.8    | 0.9    | 0      | 0      | 0      | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 968.5        |
| U2 - dose-specific alpha   |        |        |        |        |        | 966.7        |
| U3 - shared alpha          |        |        |        |        |        | 968.8        |
| U3 - dose-specific alpha   |        |        |        |        |        | 967.9        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 22                |      |      |      |      |      |        |
| DLT rate                   | 0.07 | 0.27 | 0.50 | 0.56 | 0.78 |        |
| Efficacy rate              | 0.08 | 0.25 | 0.25 | 0.50 | 0.50 |        |
| Utility 2                  | 0    | -0   | -0.2 | -0.1 | -0.3 |        |
| Utility 3                  | 42   | 44.2 | 35   | 47.6 | 38.8 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 14.1 | 53.7 | 30.3 | 1.9  | 0    | 0      |
| U2 - dose-specific alpha   | 13.7 | 54.9 | 29.2 | 2.2  | 0    | 0      |
| U3 - shared alpha          | 14   | 54.5 | 29.3 | 2.2  | 0    | 0      |
| U3 - dose-specific alpha   | 14.2 | 52.7 | 30.7 | 2.4  | 0    | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 15   | 60.2 | 12.3 | 1.3  | 0    | 11.2   |
| U2 - dose-specific alpha   | 14.6 | 60.8 | 12   | 1.4  | 0    | 11.2   |
| U3 - shared alpha          | 16.3 | 61.4 | 9.9  | 1.3  | 0    | 11.1   |
| U3 - dose-specific alpha   | 16.1 | 60.7 | 9.7  | 1.5  | 0    | 12     |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 8.4  | 13.3 | 5.3  | 1.8  | 0.1  | -      |
| U2 - dose-specific alpha   | 8.4  | 13.3 | 5.4  | 1.8  | 0.1  | -      |
| U3 - shared alpha          | 8.7  | 13.4 | 5    | 1.7  | 0.1  | -      |
| U3 - dose-specific alpha   | 8.7  | 13.4 | 5    | 1.7  | 0.1  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 2    | 5.9  | 1.5  | 0.7  | 0    | -      |
| U2 - dose-specific alpha   | 2    | 5.9  | 1.6  | 0.6  | 0    | -      |
| U3 - shared alpha          | 2.1  | 5.9  | 1.5  | 0.6  | 0    | -      |
| U3 - dose-specific alpha   | 2.1  | 5.9  | 1.5  | 0.6  | 0    | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1038.6 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1039.7 |
| U3 - shared alpha          |      |      |      |      |      | 1034.5 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1033.6 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 23                |        |        |        |        |        |              |
| DLT rate                   | 0.32   | 0.55   | 0.80   | 0.94   | 0.97   |              |
| Efficacy rate              | 0.05   | 0.05   | 0.05   | 0.05   | 0.05   |              |
| Utility 2                  | -0.3   | -0.5   | -0.8   | -0.9   | -0.9   |              |
| Utility 3                  | 30.2   | 21     | 11     | 5.4    | 4.2    |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 73.8   | 2.7    | 0      | 0      | 0      | 23.5         |
| U2 - dose-specific alpha   | 74.3   | 2.6    | 0      | 0      | 0      | 23.1         |
| U3 - shared alpha          | 72.9   | 3.3    | 0.1    | 0      | 0      | 23.7         |
| U3 - dose-specific alpha   | 73.9   | 2.8    | 0.1    | 0      | 0      | 23.2         |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 7.5    | 0.6    | 0      | 0      | 0      | 91.9         |
| U2 - dose-specific alpha   | 7.4    | 0.6    | 0      | 0      | 0      | 92           |
| U3 - shared alpha          | 7.5    | 1.1    | 0      | 0      | 0      | 91.4         |
| U3 - dose-specific alpha   | 7.2    | 0.8    | 0      | 0      | 0      | 92           |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 11.1   | 3.8    | 0.5    | 0      | 0      | -            |
| U2 - dose-specific alpha   | 11.1   | 3.9    | 0.5    | 0      | 0      | -            |
| U3 - shared alpha          | 11.1   | 3.8    | 0.5    | 0      | 0      | -            |
| U3 - dose-specific alpha   | 11.1   | 3.8    | 0.5    | 0.1    | 0      | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 2.9    | 2.1    | 0.2    | 0      | 0      | -            |
| U2 - dose-specific alpha   | 2.9    | 2.1    | 0.2    | 0      | 0      | -            |
| U3 - shared alpha          | 2.9    | 2.1    | 0.2    | 0      | 0      | -            |
| U3 - dose-specific alpha   | 2.9    | 2.1    | 0.2    | 0      | 0      | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 557.9        |
| U2 - dose-specific alpha   |        |        |        |        |        | 559          |
| U3 - shared alpha          |        |        |        |        |        | 557.2        |
| U3 - dose-specific alpha   |        |        |        |        |        | 558.3        |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 24                |      |      |      |      |      |        |
| DLT rate                   | 0.01 | 0.02 | 0.05 | 0.10 | 0.25 |        |
| Efficacy rate              | 0.05 | 0.08 | 0.20 | 0.50 | 0.50 |        |
| Utility 2                  | 0    | 0.1  | 0.1  | 0.4  | 0.2  |        |
| Utility 3                  | 42.6 | 44   | 50   | 66   | 60   |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0    | 0.1  | 0.1  | 37   | 62.8 | 0      |
| U2 - dose-specific alpha   | 0    | 0.1  | 0.1  | 36.1 | 63.7 | 0      |
| U3 - shared alpha          | 0    | 0.1  | 0.2  | 40.9 | 58.8 | 0      |
| U3 - dose-specific alpha   | 0    | 0.1  | 0.2  | 41.5 | 58.2 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 1.5  | 1    | 3    | 59.1 | 35.4 | 0      |
| U2 - dose-specific alpha   | 1.5  | 0.9  | 2.5  | 59.2 | 35.9 | 0      |
| U3 - shared alpha          | 2.7  | 1.9  | 2.7  | 62.6 | 30.1 | 0      |
| U3 - dose-specific alpha   | 2.6  | 2.1  | 4    | 62   | 29.3 | 0      |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 3.1  | 3.1  | 3.8  | 13   | 6.9  | -      |
| U2 - dose-specific alpha   | 3.1  | 3.2  | 3.8  | 13   | 7    | -      |
| U3 - shared alpha          | 3.1  | 3.2  | 3.9  | 13.6 | 6.2  | -      |
| U3 - dose-specific alpha   | 3.1  | 3.2  | 3.9  | 13.6 | 6.1  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0    | 2.4  | 1.2  | 4.5  | 2.1  | -      |
| U2 - dose-specific alpha   | 0    | 2.4  | 1.2  | 4.5  | 2.1  | -      |
| U3 - shared alpha          | 0    | 2.4  | 1.2  | 4.7  | 1.8  | -      |
| U3 - dose-specific alpha   | 0.1  | 2.4  | 1.2  | 4.7  | 1.8  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1090.8 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1090.6 |
| U3 - shared alpha          |      |      |      |      |      | 1088.3 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1087.5 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 25                |        |        |        |        |        |              |
| DLT rate                   | 0.09   | 0.18   | 0.37   | 0.63   | 0.77   |              |
| Efficacy rate              | 0.20   | 0.25   | 0.40   | 0.45   | 0.50   |              |
| Utility 2                  | 0.1    | 0.1    | 0      | -0.2   | -0.3   |              |
| Utility 3                  | 48.4   | 47.8   | 49.2   | 41.8   | 39.2   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 3.2    | 37     | 52.2   | 7.5    | 0.1    | 0            |
| U2 - dose-specific alpha   | 3.1    | 36.8   | 52.7   | 7.3    | 0.1    | 0            |
| U3 - shared alpha          | 2.7    | 36.7   | 52.7   | 7.8    | 0.1    | 0            |
| U3 - dose-specific alpha   | 3.3    | 36.6   | 52.2   | 7.8    | 0.1    | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 28.2   | 33.2   | 33     | 4.1    | 0.1    | 1.4          |
| U2 - dose-specific alpha   | 28.6   | 33     | 33.6   | 3.3    | 0.1    | 1.4          |
| U3 - shared alpha          | 32.2   | 34.2   | 29.5   | 2.8    | 0      | 1.3          |
| U3 - dose-specific alpha   | 33     | 33.7   | 29.1   | 2.9    | 0      | 1.3          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 8.8    | 9.9    | 8.5    | 2.4    | 0.2    | -            |
| U2 - dose-specific alpha   | 9      | 9.8    | 8.5    | 2.4    | 0.2    | -            |
| U3 - shared alpha          | 9.4    | 10     | 8.2    | 2.1    | 0.2    | -            |
| U3 - dose-specific alpha   | 9.4    | 10     | 8.1    | 2.1    | 0.2    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 2.2    | 4.3    | 2.4    | 0.8    | 0.1    | -            |
| U2 - dose-specific alpha   | 2.3    | 4.2    | 2.4    | 0.8    | 0.1    | -            |
| U3 - shared alpha          | 2.5    | 4.3    | 2.2    | 0.8    | 0.1    | -            |
| U3 - dose-specific alpha   | 2.5    | 4.3    | 2.2    | 0.8    | 0.1    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1109.8       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1107.9       |
| U3 - shared alpha          |        |        |        |        |        | 1103.1       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1104.5       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 26                |      |      |      |      |      |        |
| DLT rate                   | 0.15 | 0.22 | 0.30 | 0.62 | 0.80 |        |
| Efficacy rate              | 0.10 | 0.20 | 0.35 | 0.45 | 0.55 |        |
| Utility 2                  | -0.1 | -0   | 0.1  | -0.2 | -0.2 |        |
| Utility 3                  | 40   | 43.2 | 49   | 42.2 | 41   |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 14.6 | 36.9 | 38.8 | 9.1  | 0.3  | 0.3    |
| U2 - dose-specific alpha   | 15   | 36   | 38.7 | 9.7  | 0.3  | 0.3    |
| U3 - shared alpha          | 14.5 | 36.1 | 39.9 | 8.9  | 0.3  | 0.3    |
| U3 - dose-specific alpha   | 13.7 | 37.4 | 39.3 | 9    | 0.3  | 0.3    |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 20.6 | 33.2 | 28.6 | 5.2  | 0.3  | 12.1   |
| U2 - dose-specific alpha   | 19.8 | 33.5 | 28.9 | 5.1  | 0.2  | 12.5   |
| U3 - shared alpha          | 23.7 | 33.1 | 26.7 | 4.1  | 0.2  | 12.2   |
| U3 - dose-specific alpha   | 23.1 | 35.2 | 25.6 | 4    | 0.3  | 11.8   |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 8.7  | 9.2  | 7.7  | 2.7  | 0.3  | -      |
| U2 - dose-specific alpha   | 8.6  | 9.2  | 7.7  | 2.6  | 0.3  | -      |
| U3 - shared alpha          | 9.1  | 9.4  | 7.5  | 2.4  | 0.2  | -      |
| U3 - dose-specific alpha   | 9    | 9.3  | 7.5  | 2.4  | 0.2  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 2.2  | 4.3  | 2.2  | 1    | 0.1  | -      |
| U2 - dose-specific alpha   | 2.1  | 4.4  | 2.2  | 1    | 0.1  | -      |
| U3 - shared alpha          | 2.3  | 4.4  | 2.1  | 0.9  | 0.1  | -      |
| U3 - dose-specific alpha   | 2.3  | 4.4  | 2.1  | 0.9  | 0.1  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1030   |
| U2 - dose-specific alpha   |      |      |      |      |      | 1025.8 |
| U3 - shared alpha          |      |      |      |      |      | 1024.5 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1024.9 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 27                |        |        |        |        |        |              |
| DLT rate                   | 0.11   | 0.17   | 0.30   | 0.50   | 0.62   |              |
| Efficacy rate              | 0.10   | 0.20   | 0.40   | 0.42   | 0.45   |              |
| Utility 2                  | -0     | 0      | 0.1    | -0.1   | -0.2   |              |
| Utility 3                  | 41.6   | 45.2   | 52     | 45.2   | 42.2   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 4.9    | 23     | 54.5   | 16.5   | 1.1    | 0            |
| U2 - dose-specific alpha   | 4.6    | 23.7   | 53.3   | 17.5   | 0.9    | 0            |
| U3 - shared alpha          | 4.6    | 24.7   | 53.4   | 16.6   | 0.7    | 0            |
| U3 - dose-specific alpha   | 4.5    | 25.4   | 52.4   | 16.9   | 0.8    | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 14.4   | 25.7   | 45.5   | 9.3    | 0.4    | 4.7          |
| U2 - dose-specific alpha   | 14.3   | 24.5   | 46.2   | 10     | 0.3    | 4.7          |
| U3 - shared alpha          | 17.4   | 27.7   | 42.3   | 7.5    | 0.2    | 4.9          |
| U3 - dose-specific alpha   | 17.8   | 27.6   | 42.1   | 7.6    | 0.3    | 4.6          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 6.7    | 8.6    | 10     | 3.5    | 0.6    | -            |
| U2 - dose-specific alpha   | 6.7    | 8.6    | 10     | 3.5    | 0.6    | -            |
| U3 - shared alpha          | 7      | 8.8    | 9.8    | 3.2    | 0.5    | -            |
| U3 - dose-specific alpha   | 7      | 8.8    | 9.9    | 3.2    | 0.5    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 1.4    | 4.1    | 2.9    | 1.2    | 0.2    | -            |
| U2 - dose-specific alpha   | 1.5    | 4.1    | 2.9    | 1.2    | 0.2    | -            |
| U3 - shared alpha          | 1.6    | 4.2    | 2.9    | 1.1    | 0.2    | -            |
| U3 - dose-specific alpha   | 1.6    | 4.2    | 2.9    | 1.1    | 0.2    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1077.5       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1077.2       |
| U3 - shared alpha          |        |        |        |        |        | 1071.9       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1072         |

|                            |      |      |      |      |      |       |
|----------------------------|------|------|------|------|------|-------|
| Scenario 28                |      |      |      |      |      |       |
| DLT rate                   | 0.20 | 0.30 | 0.50 | 0.60 | 0.77 |       |
| Efficacy rate              | 0.05 | 0.30 | 0.32 | 0.35 | 0.40 |       |
| Utility 2                  | -0.1 | 0    | -0.2 | -0.2 | -0.4 |       |
| Utility 3                  | 35   | 46   | 39.2 | 37   | 33.2 |       |
| MTD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 48.9 | 37.8 | 10.7 | 0.8  | 0    | 1.8   |
| U2 - dose-specific alpha   | 48.6 | 37.1 | 11.6 | 0.7  | 0    | 2     |
| U3 - shared alpha          | 49.3 | 36.8 | 11.5 | 0.5  | 0    | 1.9   |
| U3 - dose-specific alpha   | 49.2 | 37   | 11.2 | 0.7  | 0    | 1.9   |
| OBD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 11.2 | 39.2 | 5.4  | 0.1  | 0    | 44.1  |
| U2 - dose-specific alpha   | 10.6 | 38.9 | 6.1  | 0.1  | 0    | 44.3  |
| U3 - shared alpha          | 11.6 | 38.7 | 5.1  | 0    | 0    | 44.6  |
| U3 - dose-specific alpha   | 11.9 | 39.2 | 4.4  | 0    | 0    | 44.5  |
| Mean administrations       |      |      |      |      |      |       |
| U2 - shared alpha          | 9.4  | 10.7 | 3.3  | 0.8  | 0.1  | -     |
| U2 - dose-specific alpha   | 9.4  | 10.6 | 3.3  | 0.9  | 0.1  | -     |
| U3 - shared alpha          | 9.6  | 10.8 | 3.2  | 0.8  | 0.1  | -     |
| U3 - dose-specific alpha   | 9.7  | 10.6 | 3.2  | 0.8  | 0.1  | -     |
| Mean IPDE administrations  |      |      |      |      |      |       |
| U2 - shared alpha          | 2.3  | 4.7  | 1    | 0.3  | 0    | -     |
| U2 - dose-specific alpha   | 2.2  | 4.6  | 1    | 0.3  | 0    | -     |
| U3 - shared alpha          | 2.3  | 4.7  | 0.9  | 0.3  | 0    | -     |
| U3 - dose-specific alpha   | 2.4  | 4.6  | 1    | 0.3  | 0    | -     |
| Mean trial duration (days) |      |      |      |      |      | Days  |
| U2 - shared alpha          |      |      |      |      |      | 880.4 |
| U2 - dose-specific alpha   |      |      |      |      |      | 881.3 |
| U3 - shared alpha          |      |      |      |      |      | 883.8 |
| U3 - dose-specific alpha   |      |      |      |      |      | 878.5 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 29                |        |        |        |        |        |              |
| DLT rate                   | 0.19   | 0.30   | 0.56   | 0.74   | 0.86   |              |
| Efficacy rate              | 0.40   | 0.40   | 0.40   | 0.40   | 0.40   |              |
| Utility 2                  | 0.2    | 0.1    | -0.2   | -0.3   | -0.5   |              |
| Utility 3                  | 56.4   | 52     | 41.6   | 34.4   | 29.6   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 31.5   | 57.4   | 8.5    | 0.3    | 0      | 2.3          |
| U2 - dose-specific alpha   | 31.3   | 57.5   | 8.8    | 0.3    | 0      | 2.1          |
| U3 - shared alpha          | 31.8   | 58.5   | 7.2    | 0.2    | 0      | 2.3          |
| U3 - dose-specific alpha   | 32.8   | 57.8   | 7.1    | 0.2    | 0      | 2.1          |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 63.7   | 31.4   | 2.1    | 0.1    | 0      | 2.7          |
| U2 - dose-specific alpha   | 64.5   | 30.1   | 2.9    | 0.1    | 0      | 2.4          |
| U3 - shared alpha          | 67.5   | 28.1   | 1.8    | 0.1    | 0      | 2.5          |
| U3 - dose-specific alpha   | 68.3   | 27.5   | 1.7    | 0.1    | 0      | 2.4          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 18.1   | 9      | 2      | 0.4    | 0      | -            |
| U2 - dose-specific alpha   | 18.1   | 8.9    | 2.1    | 0.4    | 0      | -            |
| U3 - shared alpha          | 19     | 8.6    | 1.6    | 0.3    | 0      | -            |
| U3 - dose-specific alpha   | 19     | 8.6    | 1.7    | 0.3    | 0      | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 4.4    | 3.2    | 0.6    | 0.1    | 0      | -            |
| U2 - dose-specific alpha   | 4.4    | 3.2    | 0.6    | 0.1    | 0      | -            |
| U3 - shared alpha          | 4.8    | 3.1    | 0.5    | 0.1    | 0      | -            |
| U3 - dose-specific alpha   | 4.8    | 3.1    | 0.5    | 0.1    | 0      | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1166.3       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1167.6       |
| U3 - shared alpha          |        |        |        |        |        | 1164.5       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1164.8       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 30                |      |      |      |      |      |        |
| DLT rate                   | 0.05 | 0.08 | 0.17 | 0.27 | 0.54 |        |
| Efficacy rate              | 0.12 | 0.15 | 0.20 | 0.40 | 0.50 |        |
| Utility 2                  | 0.1  | 0.1  | 0    | 0.1  | -0   |        |
| Utility 3                  | 45.2 | 45.8 | 45.2 | 53.2 | 48.4 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0.4  | 1.7  | 17.3 | 58.7 | 21.9 | 0      |
| U2 - dose-specific alpha   | 0.4  | 1.8  | 17   | 59.1 | 21.7 | 0      |
| U3 - shared alpha          | 0.4  | 2.2  | 17.6 | 59.5 | 20.3 | 0      |
| U3 - dose-specific alpha   | 0.4  | 2    | 16.6 | 60.6 | 20.4 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 16.3 | 10.8 | 15.6 | 46   | 10.9 | 0.4    |
| U2 - dose-specific alpha   | 15.7 | 11.7 | 15.4 | 45.6 | 11.2 | 0.4    |
| U3 - shared alpha          | 23.5 | 14.2 | 12.9 | 39.7 | 9.2  | 0.5    |
| U3 - dose-specific alpha   | 22.8 | 16.3 | 12.4 | 39   | 9    | 0.5    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 4.4  | 5.2  | 6.4  | 10.1 | 3.8  | -      |
| U2 - dose-specific alpha   | 4.4  | 5.2  | 6.4  | 10.2 | 3.7  | -      |
| U3 - shared alpha          | 4.7  | 5.5  | 6.4  | 10   | 3.4  | -      |
| U3 - dose-specific alpha   | 4.7  | 5.4  | 6.4  | 10   | 3.4  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.6  | 3.1  | 2    | 3.3  | 1.1  | -      |
| U2 - dose-specific alpha   | 0.6  | 3.1  | 2    | 3.4  | 1.1  | -      |
| U3 - shared alpha          | 0.7  | 3.2  | 2    | 3.3  | 1    | -      |
| U3 - dose-specific alpha   | 0.7  | 3.2  | 2    | 3.3  | 1.1  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1085.8 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1087.1 |
| U3 - shared alpha          |      |      |      |      |      | 1081.6 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1082.3 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 31                |        |        |        |        |        |              |
| DLT rate                   | 0.04   | 0.06   | 0.10   | 0.20   | 0.60   |              |
| Efficacy rate              | 0.20   | 0.50   | 0.50   | 0.50   | 0.50   |              |
| Utility 2                  | 0.2    | 0.4    | 0.4    | 0.3    | -0.1   |              |
| Utility 3                  | 50.4   | 67.6   | 66     | 62     | 46     |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 0      | 30.2   | 28.1   | 30     | 11.7   | 0            |
| U2 - dose-specific alpha   | 0      | 30     | 29.6   | 28.9   | 11.5   | 0            |
| U3 - shared alpha          | 0      | 35.7   | 28.6   | 27.6   | 8.1    | 0            |
| U3 - dose-specific alpha   | 0      | 34.8   | 28.7   | 28.1   | 8.4    | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 2.9    | 38.1   | 31.3   | 23.2   | 4.5    | 0            |
| U2 - dose-specific alpha   | 2.7    | 39.2   | 30.4   | 23.3   | 4.4    | 0            |
| U3 - shared alpha          | 4.8    | 44.8   | 28.3   | 20     | 2.1    | 0            |
| U3 - dose-specific alpha   | 4.9    | 42.6   | 29.4   | 20.6   | 2.5    | 0            |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 3.8    | 12.5   | 7.6    | 4.8    | 1.2    | -            |
| U2 - dose-specific alpha   | 3.8    | 12.6   | 7.7    | 4.7    | 1.2    | -            |
| U3 - shared alpha          | 4      | 13.6   | 7.3    | 4.3    | 0.8    | -            |
| U3 - dose-specific alpha   | 4      | 13.3   | 7.4    | 4.5    | 0.8    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 0.3    | 4.9    | 2.3    | 1.4    | 0.4    | -            |
| U2 - dose-specific alpha   | 0.3    | 4.9    | 2.3    | 1.4    | 0.4    | -            |
| U3 - shared alpha          | 0.3    | 5.2    | 2.2    | 1.3    | 0.3    | -            |
| U3 - dose-specific alpha   | 0.3    | 5.2    | 2.2    | 1.3    | 0.2    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1145.8       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1144.5       |
| U3 - shared alpha          |        |        |        |        |        | 1140.5       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1140.8       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 32                |      |      |      |      |      |        |
| DLT rate                   | 0.02 | 0.05 | 0.11 | 0.30 | 0.36 |        |
| Efficacy rate              | 0.08 | 0.15 | 0.30 | 0.30 | 0.30 |        |
| Utility 2                  | 0.1  | 0.1  | 0.2  | 0    | -0.1 |        |
| Utility 3                  | 44   | 47   | 53.6 | 46   | 43.6 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 0.1  | 0.3  | 7.6  | 42.8 | 49.2 | 0      |
| U2 - dose-specific alpha   | 0.1  | 0.4  | 8    | 42.9 | 48.6 | 0      |
| U3 - shared alpha          | 0.1  | 0.3  | 8.7  | 46.9 | 44   | 0      |
| U3 - dose-specific alpha   | 0.1  | 0.2  | 8.1  | 48.4 | 43.2 | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 11.4 | 14   | 35   | 29.1 | 10.4 | 0.1    |
| U2 - dose-specific alpha   | 11.9 | 13.4 | 34.9 | 29.2 | 10.4 | 0.2    |
| U3 - shared alpha          | 16.5 | 18.3 | 33.6 | 25.8 | 5.7  | 0.1    |
| U3 - dose-specific alpha   | 16.7 | 18   | 32.7 | 27.4 | 5.1  | 0.1    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 3.5  | 4.3  | 8.7  | 8.7  | 4.7  | -      |
| U2 - dose-specific alpha   | 3.5  | 4.3  | 8.7  | 8.7  | 4.7  | -      |
| U3 - shared alpha          | 3.6  | 4.6  | 9.1  | 8.6  | 4    | -      |
| U3 - dose-specific alpha   | 3.6  | 4.7  | 9    | 8.6  | 4.1  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.2  | 2.9  | 2.9  | 3.3  | 1.5  | -      |
| U2 - dose-specific alpha   | 0.2  | 2.9  | 2.9  | 3.3  | 1.5  | -      |
| U3 - shared alpha          | 0.3  | 3    | 3    | 3.2  | 1.3  | -      |
| U3 - dose-specific alpha   | 0.3  | 3    | 3    | 3.2  | 1.3  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1055.9 |
| U2 - dose-specific alpha   |      |      |      |      |      | 1055.2 |
| U3 - shared alpha          |      |      |      |      |      | 1052.1 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1051.7 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 33                |        |        |        |        |        |              |
| DLT rate                   | 0.01   | 0.02   | 0.09   | 0.32   | 0.38   |              |
| Efficacy rate              | 0.05   | 0.08   | 0.20   | 0.40   | 0.50   |              |
| Utility 2                  | 0      | 0.1    | 0.1    | 0.1    | 0.1    |              |
| Utility 3                  | 42.6   | 44     | 48.4   | 51.2   | 54.8   |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 0      | 0.3    | 7.9    | 52     | 39.8   | 0            |
| U2 - dose-specific alpha   | 0      | 0.1    | 7.4    | 52.9   | 39.6   | 0            |
| U3 - shared alpha          | 0      | 0.2    | 7.1    | 54.9   | 37.8   | 0            |
| U3 - dose-specific alpha   | 0      | 0.1    | 7.6    | 55.1   | 37.2   | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 8.8    | 5.9    | 11.6   | 47.3   | 26.3   | 0.1          |
| U2 - dose-specific alpha   | 8.4    | 5.8    | 11.4   | 47.7   | 26.6   | 0.1          |
| U3 - shared alpha          | 12.2   | 8.2    | 12     | 46.1   | 21.4   | 0.1          |
| U3 - dose-specific alpha   | 12.6   | 8.4    | 11.7   | 45.7   | 21.5   | 0.1          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 3.4    | 3.7    | 6      | 10.7   | 6.2    | -            |
| U2 - dose-specific alpha   | 3.3    | 3.8    | 6.1    | 10.7   | 6.2    | -            |
| U3 - shared alpha          | 3.5    | 3.8    | 6.2    | 10.9   | 5.7    | -            |
| U3 - dose-specific alpha   | 3.5    | 3.9    | 6.1    | 10.9   | 5.6    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 0.2    | 2.7    | 1.8    | 3.8    | 1.7    | -            |
| U2 - dose-specific alpha   | 0.1    | 2.7    | 1.8    | 3.8    | 1.7    | -            |
| U3 - shared alpha          | 0.2    | 2.8    | 1.9    | 3.8    | 1.5    | -            |
| U3 - dose-specific alpha   | 0.2    | 2.8    | 1.8    | 3.9    | 1.5    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1090.3       |
| U2 - dose-specific alpha   |        |        |        |        |        | 1089.2       |
| U3 - shared alpha          |        |        |        |        |        | 1087         |
| U3 - dose-specific alpha   |        |        |        |        |        | 1086.8       |

|                            |      |      |      |      |      |        |
|----------------------------|------|------|------|------|------|--------|
| Scenario 34                |      |      |      |      |      |        |
| DLT rate                   | 0.05 | 0.13 | 0.30 | 0.36 | 0.42 |        |
| Efficacy rate              | 0.05 | 0.10 | 0.30 | 0.32 | 0.35 |        |
| Utility 2                  | 0    | -0   | 0    | -0   | -0.1 |        |
| Utility 3                  | 41   | 40.8 | 46   | 44.8 | 44.2 |        |
| MTD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 1.4  | 11   | 44   | 33.6 | 10   | 0      |
| U2 - dose-specific alpha   | 1.4  | 10.8 | 44   | 33.6 | 10.2 | 0      |
| U3 - shared alpha          | 1.5  | 10.7 | 43.4 | 35.7 | 8.7  | 0      |
| U3 - dose-specific alpha   | 1.4  | 10.4 | 44   | 35.5 | 8.7  | 0      |
| OBD selection (%)          |      |      |      |      |      |        |
| U2 - shared alpha          | 12.7 | 13.2 | 47.5 | 19.2 | 3.1  | 4.3    |
| U2 - dose-specific alpha   | 13.2 | 12.9 | 47.7 | 19   | 3.2  | 4      |
| U3 - shared alpha          | 15.8 | 16   | 44.1 | 17.9 | 2.2  | 4      |
| U3 - dose-specific alpha   | 16.2 | 15.9 | 44.2 | 17.6 | 2    | 4.1    |
| Mean administrations       |      |      |      |      |      |        |
| U2 - shared alpha          | 5.1  | 6.7  | 9.9  | 5.9  | 2.1  | -      |
| U2 - dose-specific alpha   | 5.1  | 6.6  | 9.9  | 5.9  | 2.1  | -      |
| U3 - shared alpha          | 5.5  | 6.8  | 9.9  | 5.8  | 1.7  | -      |
| U3 - dose-specific alpha   | 5.5  | 6.8  | 9.9  | 5.7  | 1.9  | -      |
| Mean IPDE administrations  |      |      |      |      |      |        |
| U2 - shared alpha          | 0.9  | 3.7  | 3.1  | 2.1  | 0.7  | -      |
| U2 - dose-specific alpha   | 0.9  | 3.7  | 3.1  | 2.1  | 0.7  | -      |
| U3 - shared alpha          | 1.1  | 3.8  | 3.1  | 2    | 0.6  | -      |
| U3 - dose-specific alpha   | 1.1  | 3.8  | 3.1  | 2    | 0.6  | -      |
| Mean trial duration (days) |      |      |      |      |      | Days   |
| U2 - shared alpha          |      |      |      |      |      | 1054   |
| U2 - dose-specific alpha   |      |      |      |      |      | 1053.7 |
| U3 - shared alpha          |      |      |      |      |      | 1050.1 |
| U3 - dose-specific alpha   |      |      |      |      |      | 1052   |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 35                |        |        |        |        |        |              |
| DLT rate                   | 0.03   | 0.09   | 0.30   | 0.40   | 0.45   |              |
| Efficacy rate              | 0.20   | 0.45   | 0.45   | 0.45   | 0.45   |              |
| Utility 2                  | 0.2    | 0.4    | 0.1    | 0.1    | 0      |              |
| Utility 3                  | 50.8   | 63.4   | 55     | 51     | 49     |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 0.1    | 23.7   | 48.9   | 23     | 4.3    | 0            |
| U2 - dose-specific alpha   | 0.1    | 23.5   | 47.8   | 23.4   | 5.2    | 0            |
| U3 - shared alpha          | 0.1    | 28.2   | 46.3   | 21.7   | 3.7    | 0            |
| U3 - dose-specific alpha   | 0.1    | 27.4   | 47.9   | 20.6   | 4      | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 7.3    | 43.9   | 35.2   | 11.5   | 2      | 0.1          |
| U2 - dose-specific alpha   | 7.7    | 42.1   | 34.9   | 12.7   | 2.5    | 0.1          |
| U3 - shared alpha          | 11     | 46.5   | 30.8   | 9.7    | 1.9    | 0.1          |
| U3 - dose-specific alpha   | 11.1   | 47.7   | 31     | 8.4    | 1.7    | 0.1          |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 4.4    | 12.3   | 8.6    | 3.7    | 0.9    | -            |
| U2 - dose-specific alpha   | 4.4    | 12.1   | 8.6    | 3.8    | 1      | -            |
| U3 - shared alpha          | 4.6    | 13.2   | 8      | 3.4    | 0.8    | -            |
| U3 - dose-specific alpha   | 4.7    | 13.1   | 8      | 3.3    | 0.8    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 0.5    | 4.8    | 2.4    | 1.2    | 0.3    | -            |
| U2 - dose-specific alpha   | 0.5    | 4.8    | 2.4    | 1.2    | 0.3    | -            |
| U3 - shared alpha          | 0.6    | 5.2    | 2.2    | 1      | 0.2    | -            |
| U3 - dose-specific alpha   | 0.6    | 5.2    | 2.2    | 1      | 0.2    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 1149         |
| U2 - dose-specific alpha   |        |        |        |        |        | 1148.4       |
| U3 - shared alpha          |        |        |        |        |        | 1142.2       |
| U3 - dose-specific alpha   |        |        |        |        |        | 1142.4       |

|                            |      |      |      |      |      |       |
|----------------------------|------|------|------|------|------|-------|
| Scenario 36                |      |      |      |      |      |       |
| DLT rate                   | 0.13 | 0.29 | 0.35 | 0.40 | 0.43 |       |
| Efficacy rate              | 0.05 | 0.20 | 0.22 | 0.30 | 0.35 |       |
| Utility 2                  | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 |       |
| Utility 3                  | 37.8 | 40.4 | 39.2 | 42   | 43.8 |       |
| MTD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 24.9 | 34.4 | 27   | 10.4 | 3.1  | 0.2   |
| U2 - dose-specific alpha   | 24.8 | 33.9 | 27.7 | 10.5 | 2.9  | 0.2   |
| U3 - shared alpha          | 23.9 | 36.4 | 26.7 | 10   | 2.8  | 0.2   |
| U3 - dose-specific alpha   | 24.2 | 36.1 | 26.4 | 10.2 | 2.9  | 0.2   |
| OBD selection (%)          |      |      |      |      |      |       |
| U2 - shared alpha          | 13.4 | 41.4 | 15.4 | 5.1  | 1    | 23.7  |
| U2 - dose-specific alpha   | 14.3 | 40.6 | 16.4 | 4.6  | 1    | 23.1  |
| U3 - shared alpha          | 16.8 | 42.4 | 13.5 | 4    | 0.5  | 22.8  |
| U3 - dose-specific alpha   | 15.8 | 42.6 | 13.3 | 4.5  | 0.6  | 23.2  |
| Mean administrations       |      |      |      |      |      |       |
| U2 - shared alpha          | 8.1  | 10.2 | 5.3  | 2.9  | 0.8  | -     |
| U2 - dose-specific alpha   | 8.1  | 10.2 | 5.3  | 2.9  | 0.8  | -     |
| U3 - shared alpha          | 8.4  | 10.3 | 5.1  | 2.7  | 0.7  | -     |
| U3 - dose-specific alpha   | 8.4  | 10.4 | 5.1  | 2.7  | 0.7  | -     |
| Mean IPDE administrations  |      |      |      |      |      |       |
| U2 - shared alpha          | 1.9  | 4.9  | 1.7  | 1.1  | 0.3  | -     |
| U2 - dose-specific alpha   | 1.8  | 4.9  | 1.7  | 1.1  | 0.3  | -     |
| U3 - shared alpha          | 2    | 5    | 1.6  | 1    | 0.2  | -     |
| U3 - dose-specific alpha   | 2    | 5    | 1.6  | 1    | 0.2  | -     |
| Mean trial duration (days) |      |      |      |      |      | Days  |
| U2 - shared alpha          |      |      |      |      |      | 958.5 |
| U2 - dose-specific alpha   |      |      |      |      |      | 959.2 |
| U3 - shared alpha          |      |      |      |      |      | 957   |
| U3 - dose-specific alpha   |      |      |      |      |      | 955.6 |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No MTD/OBD % |
|----------------------------|--------|--------|--------|--------|--------|--------------|
| Scenario 37                |        |        |        |        |        |              |
| DLT rate                   | 0.08   | 0.30   | 0.40   | 0.45   | 0.50   |              |
| Efficacy rate              | 0.05   | 0.20   | 0.20   | 0.20   | 0.20   |              |
| Utility 2                  | -0     | -0.1   | -0.2   | -0.2   | -0.3   |              |
| Utility 3                  | 39.8   | 40     | 36     | 34     | 32     |              |
| MTD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 19.8   | 43.4   | 28.9   | 6.5    | 1.4    | 0            |
| U2 - dose-specific alpha   | 19.5   | 43.9   | 28.8   | 6.5    | 1.3    | 0            |
| U3 - shared alpha          | 18.9   | 43.6   | 29     | 7.2    | 1.3    | 0            |
| U3 - dose-specific alpha   | 19     | 43.8   | 28.9   | 7.1    | 1.2    | 0            |
| OBD selection (%)          |        |        |        |        |        |              |
| U2 - shared alpha          | 13.4   | 50.2   | 13.7   | 2.5    | 0.4    | 19.8         |
| U2 - dose-specific alpha   | 13     | 50.2   | 13.5   | 2.8    | 0.4    | 20.1         |
| U3 - shared alpha          | 16.1   | 50.3   | 11.6   | 2.4    | 0.2    | 19.4         |
| U3 - dose-specific alpha   | 16.1   | 50.9   | 11.3   | 2      | 0.2    | 19.5         |
| Mean administrations       |        |        |        |        |        |              |
| U2 - shared alpha          | 7.9    | 11.6   | 5.4    | 2.5    | 0.7    | -            |
| U2 - dose-specific alpha   | 8      | 11.5   | 5.3    | 2.5    | 0.7    | -            |
| U3 - shared alpha          | 8.3    | 11.5   | 5.2    | 2.4    | 0.6    | -            |
| U3 - dose-specific alpha   | 8.3    | 11.5   | 5.2    | 2.4    | 0.6    | -            |
| Mean IPDE administrations  |        |        |        |        |        |              |
| U2 - shared alpha          | 1.8    | 5.5    | 1.7    | 1      | 0.2    | -            |
| U2 - dose-specific alpha   | 1.8    | 5.5    | 1.7    | 1      | 0.2    | -            |
| U3 - shared alpha          | 2      | 5.5    | 1.6    | 0.9    | 0.2    | -            |
| U3 - dose-specific alpha   | 2      | 5.5    | 1.6    | 0.9    | 0.2    | -            |
| Mean trial duration (days) |        |        |        |        |        | Days         |
| U2 - shared alpha          |        |        |        |        |        | 979.6        |
| U2 - dose-specific alpha   |        |        |        |        |        | 977.4        |
| U3 - shared alpha          |        |        |        |        |        | 975.8        |
| U3 - dose-specific alpha   |        |        |        |        |        | 976.4        |

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.